



Prediction, Estimation, and Attribution

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ABSTRACT

The scientific needs and computational limitations of the twentieth century fashioned classical statistical methodology. Both the needs and limitations have changed in the twenty-first, and so has the methodology. Large-scale prediction algorithms—neural nets, deep learning, boosting, support vector machines, random forests—have achieved star status in the popular press. They are recognizable as heirs to the regression tradition, but ones carried out at enormous scale and on titanic datasets. How do these algorithms compare with standard regression techniques such as ordinary least squares or logistic regression? Several key discrepancies will be examined, centering on the differences between prediction and estimation or prediction and attribution (significance testing). Most of the discussion is carried out through small numerical examples.

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1. Introduction

Statistical regression methods go back to Gauss and Legendre in the early 1800s, and especially to Galton in 1877. During the twentieth century, regression ideas were adapted to a variety of important statistical tasks: the prediction of new cases, the estimation of regression surfaces, and the assignment of significance to individual predictors, what I have called “attribution” in the title of this article. Many of the most powerful ideas of twentieth century statistics are connected with regression: least squares fitting, logistic regression, generalized linear models, ANOVA, predictor significance testing, regression to the mean.

The twenty-first century has seen the rise of a new breed of what can be called “pure prediction algorithms”—neural nets, deep learning, boosting, support vector machines, random forests—recognizably in the Gauss–Galton tradition, but able to operate at immense scales, with millions of data points and even more millions of predictor variables. Highly successful at automating tasks like online shopping, machine translation, and airline information, the algorithms (particularly deep learning) have become media darlings in their own right, generating an immense rush of interest in the business world. More recently, the rush has extended into the world of science, a one-minute browser search producing “deep learning in biology”; “computational linguistics and deep learning”; and “deep learning for regulatory genomics.”

How do the pure prediction algorithms relate to traditional regression methods? That is the central question pursued in what follows. A series of salient differences will be examined—differences of assumption, scientific philosophy, and goals. The story is a complicated one, with no clear winners or losers; but a rough summary, at least in my mind, is that the pure prediction

algorithms are a powerful addition to the statistician’s armory, yet substantial further development is needed for their routine scientific applicability. Such development is going on already in the statistical world, and has provided a welcome shot of energy into our discipline.

This article, originally a talk, is written with a broad brush, and is meant to be descriptive of current practice rather than normative of how things have to be. No previous knowledge of the various prediction algorithms is assumed, though that will sorely underestimate many readers.

This is not a research paper, and most of the argumentation is carried out through numerical examples. These are of small size, even miniscule by current prediction standards. A certain gigantism has gripped the prediction literature, with swelling prefixes such as *tera-*, *peta-*, and *exabestowing* bragging rights. But small datasets can be better for exposing the limitations of a new methodology.

An excellent reference for prediction methods, both traditional and modern, is Hastie, Tibshirani, and Friedman (2009). Very little will be said here about the mechanics of the pure prediction algorithms: just enough, I hope, to get across the idea of how radically different they are from their traditional counterparts.

2. Surface Plus Noise Models

For both the prediction algorithms and traditional regression methods, we will assume that the data $\mathbf{d}$ available to the statistician has this structure:

$$\mathbf{d} = \{(x_i, y_i), \quad i = 1, 2, \dots, n\}; \quad (1)$$

here x_i is a p -dimensional vector of predictors taking its value in a known space $\mathcal{X}$ contained in $\mathbb{R}^p$, and y_i is a real-valued

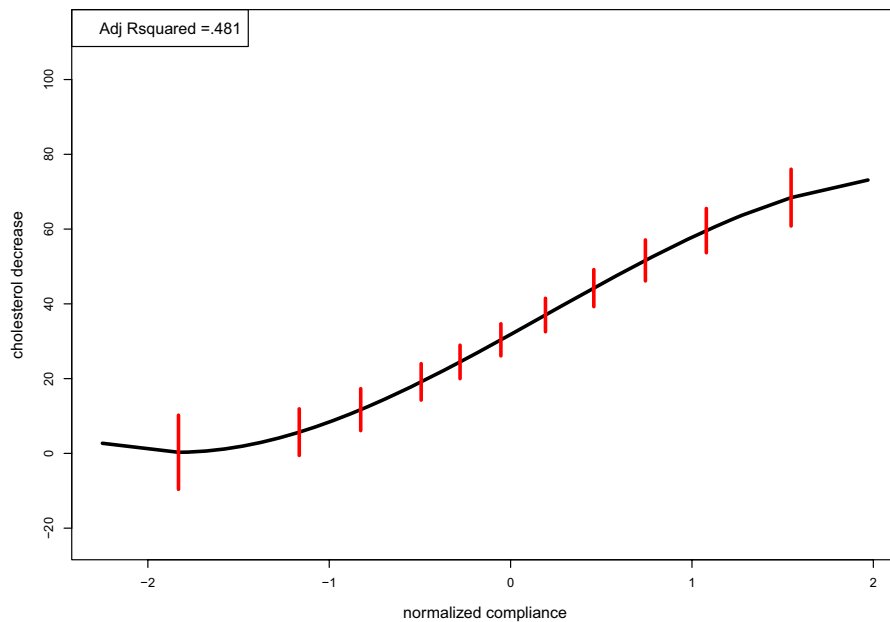


Figure 1. Black curve is OLS fitted regression to cholestyramine data (dots); vertical bars indicate $\pm$ one standard error estimation.

response. The n pairs are assumed to be independent of each other. More concisely we can write

$$\mathbf{d} = \{\mathbf{x}, \mathbf{y}\}, \quad (2)$$

where $\mathbf{x}$ is the $n \times p$ matrix having x_i^t as its i th row, and $\mathbf{y} = (y_1, y_2, \dots, y_n)^t$. Perhaps the most traditional of traditional regression models is

$$y_i = x_i^t \beta + \epsilon_i \quad (i = 1, 2, \dots, n), \quad (3)$$

$\epsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, that is, “ordinary least squares with normal errors.” Here, β is an unknown p -dimensional parameter vector. In matrix notation,

$$\mathbf{y} = \mathbf{x}\beta + \boldsymbol{\epsilon}. \quad (4)$$

For any choice of x in $\mathcal{X}$, model (3) says that the response y has expectation $\mu = x^t \beta$, so $y \sim \mathcal{N}(\mu, \sigma^2)$. The linear surface $\mathcal{S}_\beta$,

$$\mathcal{S}_\beta = \{\mu = x^t \beta, \quad x \in \mathcal{X}\}, \quad (5)$$

contains all the true expectations, but the truth is blurred by the noise terms ϵ_i .

More generally, we might expand (3) to

$$y_i = s(x_i, \beta) + \epsilon_i \quad (i = 1, 2, \dots, n), \quad (6)$$

where $s(x, \beta)$ is some known functional form that, for any fixed value of β , gives expectation $\mu = s(x, \beta)$ as a function of $x \in \mathcal{X}$. Now the surface of true expectations, that is, the *regression surface*, is

$$\mathcal{S}_\beta = \{\mu = s(x, \beta), \quad x \in \mathcal{X}\}. \quad (7)$$

Most traditional regression methods depend on some sort of *surface plus noise* formulation (though “plus” may refer to, say, binomial variability). The surface describes the scientific truths we wish to learn, but we can only observe points on the surface obscured by noise. The statistician’s traditional estimation task

is to learn as much as possible about the surface from the data $\mathbf{d}$.

Figure 1 shows a small example, taken from a larger dataset in Efron and Feldman (1991): $n = 164$ male doctors volunteered to take the cholesterol-lowering drug cholestyramine. Two numbers were recorded for each doctor,

$$\begin{aligned} x_i &= \text{normalized compliance} \quad \text{and} \\ y_i &= \text{observed cholesterol decrease}. \end{aligned} \quad (8)$$

Compliance, the proportion of the intended dose actually taken, ranged from 0% to 100%, -2.25 to 1.97 on the normalized scale, and of course it was hoped to see larger cholesterol decreases for the better compliers.

A normal regression model (6) was fit, with

$$s(x_i, \beta) = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3, \quad (9)$$

in other words, a cubic regression model. The black curve is the estimated surface

$$\hat{\mathcal{S}} = \left\{ s(x, \hat{\beta}) \quad \text{for } x \in \mathcal{X} \right\}, \quad (10)$$

fit by maximum likelihood or, equivalently, by ordinary least squares (OLS). The vertical bars indicate $\pm$ one standard error for the estimated values $s(x, \hat{\beta})$, at 11 choices of x , showing how inaccurate $\hat{\mathcal{S}}$ might be as an estimate of the true $\mathcal{S}$.

That is the estimation side of the story. As far as attribution is concerned, only $\hat{\beta}_0$ and $\hat{\beta}_1$ were significantly nonzero. The adjusted R^2 was 0.482, a traditional measure of the model’s predictive power.

Another mainstay of traditional methodology is logistic regression. Table 1 concerns the *neonate data* (Mediratta et al. 2019): $n = 812$ very sick babies at an African facility were observed over the course of one year, 605 who lived and 207 who died. Eleven covariates were measured at entry: gestational age, body weight, Apgar score, etc., so x_i in (1) was 11-dimensional, while y_i equaled 0 or 1 as the baby lived or died. This is a surface

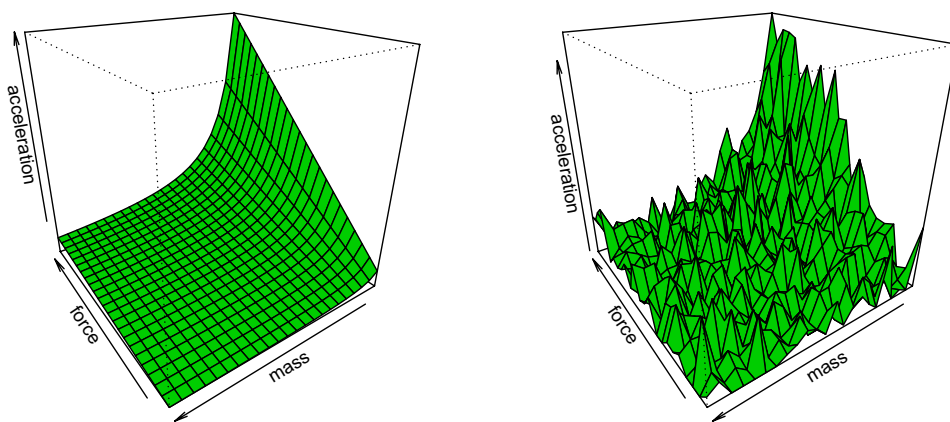


Figure 2. On left, a surface depicting Newton's second law of motion, $\text{acceleration} = \text{force}/\text{mass}$; on right, a noisy version.

Table 1. Logistic regression analysis of neonate data.

	Estimate	SE	<i>p</i> -value
Intercept	−1.549	0.457	0.001***
gest	−0.474	0.163	0.004**
ap	−0.583	0.110	0.000***
bwei	−0.488	0.163	0.003**
resp	0.784	0.140	0.000***
cpap	0.271	0.122	0.027*
ment	1.105	0.271	0.000***
rate	−0.089	0.176	0.612
hr	0.013	0.108	0.905
head	0.103	0.111	0.355
gen	−0.001	0.109	0.994
temp	0.015	0.124	0.905

NOTE: Significant two-sided *p*-values indicated for 6 of 11 predictors; estimated logistic regression made 18% prediction errors.

plus noise model, with a linear logistic surface and Bernoulli noise.

The 11 predictor variables were standardized to have mean 0 and variance 1, after which logistic regression analysis was carried out. Table 1 shows some of the output. Columns 1 and 2 give estimates and standard errors for the regression coefficients (which amount to a description of the estimated linear logistic surface $\hat{S}$ and its accuracy).

Column 3 shows standard two-sided *p*-values for the 11 variables, 6 of which are significantly nonzero, 5 of them strongly so. This is the attribution part of the analysis. As far as prediction is concerned, the fitted logistic regression model gave an estimated probability p_i of death for each baby. The prediction rule

$$\begin{array}{ll} \text{if } p_i > 0.25 & \text{predict dies} \\ p_i \leq 0.25 & \text{predict lives} \end{array} \quad (11)$$

had an empirical error rate of 18%. (Threshold 0.25 was chosen to compensate for the smaller proportion of deaths.)

All of this is familiar stuff, serving here as a reminder of how traditional regression analyses typically begin: a description of the underlying scientific truth (the “surface”) is formulated, along with a model of the errors that obscure direct observation. The pure prediction algorithms follow a different path, as described in Section 3.

The left panel of Figure 2 shows the surface representation of a scientific icon, Newton's second law of motion,

$$\text{acceleration} = \frac{\text{force}}{\text{mass}}. \quad (12)$$

It is pleasing to imagine the second law falling full-born out of Newton's head, but he was a master of experimentation. The right panel shows a (fanciful) picture of what experimental data might have looked like.¹

In the absence of genius-level insight, statistical estimation theory is intended as an instrument for peering through the noisy data and discerning a smooth underlying truth. Neither the cholestyramine nor the neonate examples is as fundamental as Newton's second law but they share the goal of extracting dependable scientific structure in a noisy environment. The noise is ephemeral but the structure, hopefully, is eternal, or at least long-lasting (see Section 8).

3. The Pure Prediction Algorithms

The twenty-first century² has seen the rise of an extraordinary collection of prediction algorithms: *random forests*, *gradient boosting*, *support vector machines*, *neural nets* (including *deep learning*), and others. I will refer to these collectively as the “pure prediction algorithms” to differentiate them from the traditional prediction methods illustrated in the previous section. Some spectacular successes—machine translation, iPhone's Siri, facial recognition, championship chess, and Go programs—have elicited a tsunami of public interest. If media attention is the appropriate metric, then the pure prediction algorithms are our era's statistical stars.

The adjective “pure” is justified by the algorithms' focus on prediction, to the neglect of estimation and attribution. Their basic strategy is simple: to go directly for high predictive accuracy and not worry about surface plus noise models. This has some striking advantages and some drawbacks, too. Both advantages and drawbacks will be illustrated in what follows.

A prediction algorithm is a general program for inputting a dataset $\mathbf{d} = \{(x_i, y_i), i = 1, 2, \dots, n\}$ (1) and outputting a rule

¹ A half-century earlier, Galileo famously used inclined planes and a water clock to estimate the acceleration of falling objects.

² Actually, the “long twenty-first century,” much of the activity beginning in the 1990s.

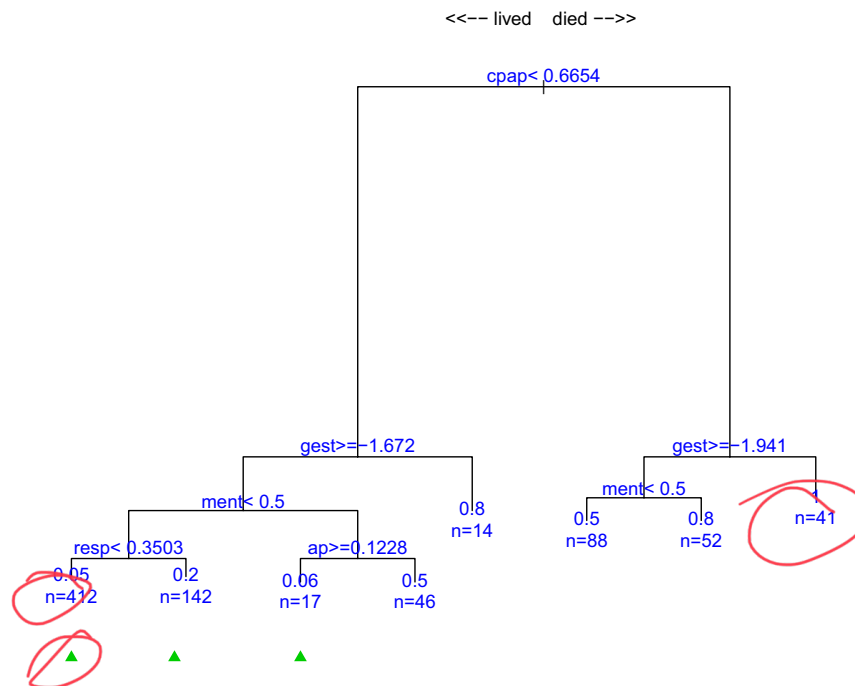


Figure 3. Classification tree for neonate data. Triangled terminal nodes predict baby lives, circled predict baby dies; the rule has apparent prediction error rate 17% and cross-validated rate 18%.

$f(x, \mathbf{d})$ that, for any predictor vector x , yields a prediction

$$\hat{y} = f(x, \mathbf{d}). \quad (13)$$

We hope that the apparent error rate of the rule, for classification problems the proportion of cases where $\hat{y}_i \neq y_i$,

$$\widehat{\text{err}} = \#\{f(x_i, \mathbf{d}) \neq y_i\}/n \quad (14)$$

is small. More crucially, we hope that the true error rate

$$\text{Err} = E\{f(X, \mathbf{d}) \neq Y\} \quad (15)$$

is small, where (X, Y) is a random draw from whatever probability distribution gave the (x_i, y_i) pairs in $\mathbf{d}$; see Section 6. Random forests, boosting, deep learning, etc. are algorithms that have well-earned reputations for giving small values of Err in complicated situations.

Besides being very different from traditional prediction methods, the pure prediction algorithms are very different from each other. The least intricate and easiest to describe is *random forests* (Breiman 2001). For dichotomous prediction problems, like that for the neonate babies, random forests depends on ensembles of *classification trees*.

Figure 3 shows a single classification tree obtained by applying the R program `Rpart3`³ to the neonates. At the top of the tree all 812 babies were divided into two groups: those with *cpap* (an airway blockage signifier) less than threshold 0.6654 were put into the more-favorable prognosis group to the left; those with *cpap* ≥ 0.6654 were shunted right into the less-favorable prognosis group. The predictor *cpap* and threshold 0.6654 were chosen to maximize, among all possible (predictor, threshold) choices, the difference in observed death rates between the

two groups.⁴ Next, each of the two groups was itself split in two, following the same Gini criterion. The splitting process continued until certain stopping rules were invoked, involving very small or very homogeneous groups.

At the bottom of Figure 3, the splitting process ended at eight *terminal nodes*: the node at far left contained 412 of the original 812 babies, only 5% of which were deaths; the far right node contained 41 babies, all of which were deaths. Triangles indicate the three terminal nodes having death proportions less than the original proportion 25.5%, while circles indicate proportions exceeding 25.5%. The prediction rule is “lives” at triangles, “dies” at circles. If a new baby arrived at the facility with vector x of 11 measurements, the doctors could predict life or death by following x down the tree to termination.

This prediction rule has apparent error rate 17%, taking the observed node proportions, 0.05, etc., as true. Classification trees have a reputation for being greedy overfitters, but in this case a 10-fold cross-validation analysis gave error rate 18%, nearly the same. The careful “traditional” analysis of the neonate data in Mediratta et al. (2019) gave a cross-validated error rate of 20%. It is worth noting that the splitting variables in Figure 3 agree nicely with those found significant in Table 1.

So far so good for regression trees, but with larger examples they have earned a reputation for poor predictive performance; see Section 9.2 of Breiman (2001). As an improvement, Breiman’s *random forest* algorithm relies on averaging a large number of bootstrap trees, each generated as follows:

⁴More precisely: if n_L and n_R are the numbers in the left and right groups, and $\hat{p}_L$ and $\hat{p}_R$ the proportions of deaths, then the algorithm minimized the *Gini criterion* $n_L\hat{p}_L(1-\hat{p}_L) + n_R\hat{p}_R(1-\hat{p}_R)$. This equals $n\hat{p}(1-\hat{p}) - (n_L n_R / n)(\hat{p}_L - \hat{p}_R)^2$, where $n = n_L + n_R$ and $\hat{p} = (n_L\hat{p}_L + n_R\hat{p}_R)/n$, so that minimizing Gini’s criterion is equivalent to maximizing $(\hat{p}_L - \hat{p}_R)^2$, for any given values of n_L and n_R .

³A knockoff of CART (Breiman et al. 1984).

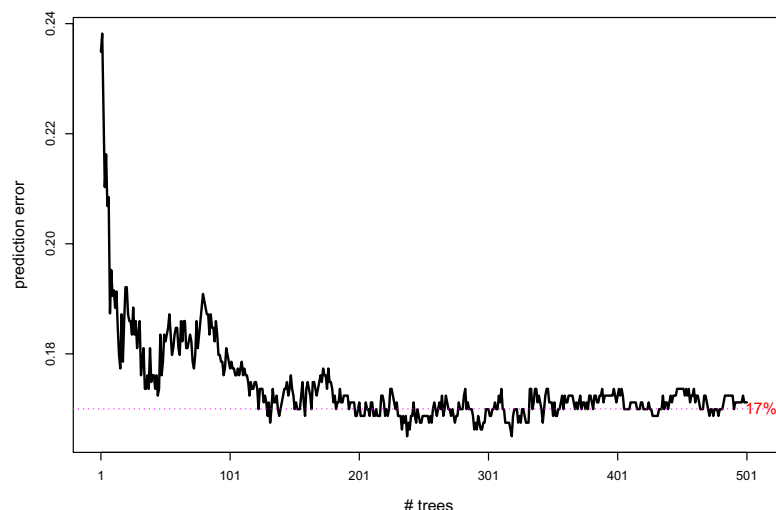


Figure 4. Random forest prediction error rate for neonate data, as a function of number of bootstrapped trees; it has cross-validated error rate 17%.

1. Draw a nonparametric bootstrap sample $\mathbf{d}^*$ from the original data $\mathbf{d}$, that is, a random sample of n pairs (x_i, y_i) chosen *with* replacement from $\mathbf{d}$.
2. Construct a classification tree from $\mathbf{d}^*$ as before, but choose each split using only a random subset of p^* predictors chosen independently from the original p ($p^* \doteq \sqrt{p}$).

Having generated, say, B such classification trees, a newly observed x is classified by following it down to termination in each tree; finally, $\hat{y} = f(x, \mathbf{d})$ is determined by the majority of the B votes. Typically, B is taken in the range 100–1000.

Random forests was applied to the neonate prediction problem, using the R program `randomForest`, with the results graphed in Figure 4. The prediction error rate⁵ is shown as a function of the number of bootstrap trees sampled. In all, $B = 501$ trees were used but there was not much change after 200. The overall prediction error rate fluctuated around 17%, only a small improvement over the 18% cross-validated rate in Figure 3. Random forests is shown to better advantage in the microarray example of Section 4.

Random forests begins with the p columns of x as predictors, but then coins a host of new predictors via the splitting process (e.g., “cpap less than or greater than 0.6654”). The new variables bring a high degree of interaction to the analysis, for instance, between cpap and gest in Figure 3. Though carried out differently, high interactivity and fecund coinage of predictor variables are hallmarks of all pure prediction algorithms.

4. A Microarray Prediction Problem

Newsworthy breakthroughs for pure prediction algorithms have involved truly enormous datasets. The original English/French translator tool on Google, for instance, was trained on millions of parallel snippets of English and French obtained from Canadian and European Union legislative records. There is nothing of that size to offer here but, as a small step up from the neonate data, we will consider a microarray study of prostate cancer.

Table 2. Number of random forest test set errors in 100 random training/test splits of prostate data.

Errors	0	1	2	3	4	5	7
Frequency	3	26	39	12	5	4	1

The study involved $n = 102$ men, 52 cancer patients and 50 normal controls. Each man’s genetic expression levels were measured on a panel of $p = 6033$ genes,

$$x_{ij} = \text{activity of } j\text{th gene for } i\text{th man}, \quad (16)$$

$i = 1, 2, \dots, 102$ and $j = 1, 2, \dots, 6033$. The $n \times p$ matrix $\mathbf{x}$ is much wider than it is tall in this case, “wide data” being the trendy name for $p \gg n$ situations, as contrasted with the $p \ll n$ “tall” datasets traditionally favored.

Random forests was put to the task of predicting *normal* or *cancer* from a man’s microarray measurements. Following standard procedure, the 102 men were randomly divided into *training* and *test* sets of size 51,⁶ each having 25 normal controls and 26 cancer patients.

The training data $\mathbf{d}_{\text{train}}$ consists of 51 (x, y) pairs, x a vector of $p = 6033$ genetic activity measurements and y equal 0 or 1 indicating a normal or cancer patient. R program `randomForest` yielded prediction rule $f(x, \mathbf{d}_{\text{train}})$. This rule was applied to the test set, yielding predictions $\hat{y}_i = f(x_i, \mathbf{d}_{\text{train}})$ for the 51 test subjects.

Figure 5 graphs the test set error rate as the number of random forest trees increased. After 100 trees, the test set error rate was 2%. That is, $\hat{y}_i$ agreed with y_i , the actual outcome, for 50 of the 51 test set subjects: an excellent performance by anyone’s standards! This was not a particularly lucky result. Subsequently, random training/test set splits were carried out 100 times, each time repeating the random forest calculations in Figure 5 and counting the number of test set errors. The modal number of errors was 2, as seen in Table 2, with “1 prediction error” occurring frequently.

⁵These are “out-of-bag” estimates of prediction error, a form of cross-validation explained in Appendix A.

⁶It would be more common to choose, say, 81 training and 21 test, but for the comparisons that follow it will be helpful to have larger test sets.

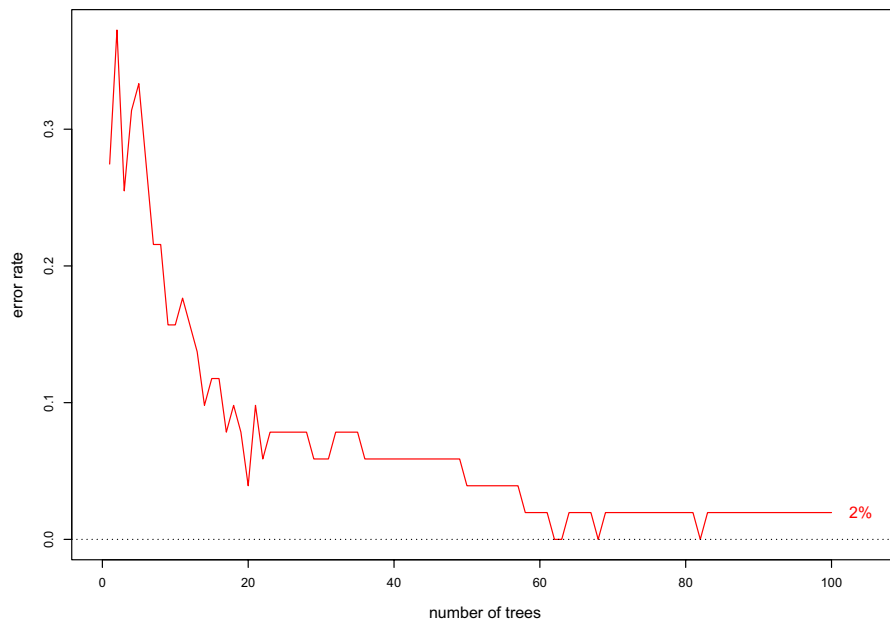


Figure 5. Test set error rate for random forests applied to prostate cancer microarray study, as a function of number of bootstrap trees.

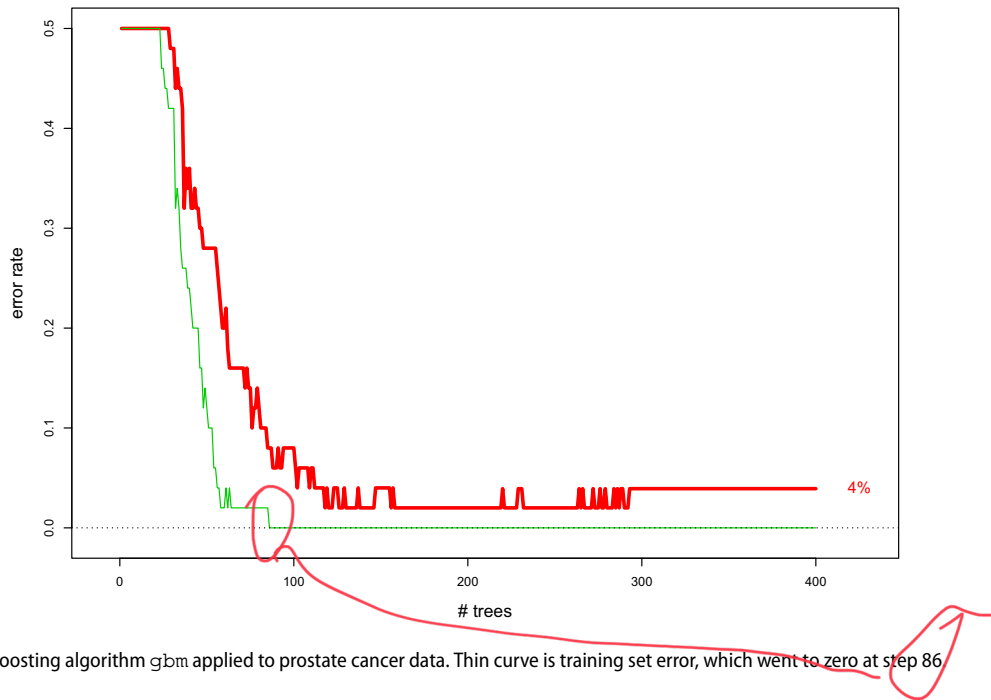


Figure 6. Test set error for boosting algorithm gbm applied to prostate cancer data. Thin curve is training set error, which went to zero at step 86

A classification tree can be thought of as a function $f(x)$ taking values 0 or 1 for x in its sample space $\mathcal{X}$. The tree in Figure 3 partitions the 11-dimensional space $\mathcal{X}$ into 8 rectangular regions, three of which having $y = 0$ and five having $y = 1$. A simpler function is obtained by stopping the division process after the first split, in which case $\mathcal{X}$ is divided into just two regions, $\text{cpap} < 0.6654$ and $\text{cpap} \geq 0.6654$. Such simple trees are picturesquely known as “stumps.”

This brings up another prominent pure prediction method, *boosting*. Figure 6 shows the results of applying the R program gbm (for gradient boosting modeling) to the prostate cancer prediction problem.⁷ Gbm sequentially fits a weighted sum of

classification trees,

$$\sum_{k=1}^K w_k f_k(x), \quad (17)$$

at step $k + 1$ choosing tree $f_{k+1}(x)$ to best improve the fit. The weights w_k are kept small to avoid getting trapped in a bad sequence. After 400 steps, Figure 6 shows a test sample error of 4%, that is, two mistakes out of 51; once again, an impressive performance. (The examples in Hastie, Tibshirani, and Friedman (2009) show gbm usually doing a little better than random forests.)

In the evocative language of boosting, the stumps going into Figure 6’s construction are called “weak learners”: any one of them by itself would barely lower prediction errors below 50%.

⁷Applied with $d = 1$, that is, fitting stumps, and shrinkage factor 0.1.

That a myriad of weak learners can combine so effectively is a happy surprise and a central advance of the pure prediction enterprise. In contrast, traditional methods focus on strong individual predictors, as with the asterisks in Table 1, a key difference to be discussed in subsequent sections.

The light curve in Figure 6 traces the gbm rule's error rate on its own training set. It went to zero at step 86 but training continued on, with some improvement in test error. Cross-validation calculations give some hint of when to stop the fitting process—here we would have done better to stop at step 200—but it's not a settled question.

The umbrella package *keras* was used to apply neural nets/deep learning to the prostate data. Results were poorer than for random forests or gbm: 7 or 8 errors on the test set depending on the exact stopping rule. A support vector machine algorithm did worse still, with 11 test set errors.

The deep learning algorithm is much more intricate than the others, reporting “780,736 parameters used,” these being internally adjusted *tuning parameters* set by cross-validation. That this is possible at all is a tribute to modern computing power, the underlying enabler of the pure prediction movement.

5. Advantages and Disadvantages of Prediction

For those of us who have struggled to find “significant” genes in a microarray study,⁸ the almost perfect prostate cancer predictions of random forests and gbm have to come as a disconcerting surprise. Without discounting the surprise, or the ingenuity of the prediction algorithms, a contributing factor might be that prediction is an easier task than either attribution or estimation. This is a difficult suspicion to support in general, but a couple of examples help make the point.

Regarding estimation, suppose that we observe 25 independent replications from a normal distribution with unknown expectation μ ,

$$x_1, x_2, \dots, x_{25} \stackrel{\text{ind}}{\sim} \mathcal{N}(\mu, 1), \quad (18)$$

and consider estimating μ with either the sample mean $\bar{x}$ or the sample median $\check{x}$. As far as squared error is concerned, the mean is an overwhelming winner, being more than half again more efficient,

$$E\{(\check{x} - \mu)^2\}/E\{(\bar{x} - \mu)^2\} \doteq 1.57. \quad (19)$$

Suppose instead that the task is to predict the value of a new, independent realization $X \sim \mathcal{N}(\mu, 1)$. The mean still wins, but now by only 2%,

$$E\{(X - \check{x})^2\}/E\{(X - \bar{x})^2\} = 1.02. \quad (20)$$

The reason, of course, is that most of the prediction error comes from the variability of X , which neither $\bar{x}$ nor $\check{x}$ can cure.⁹

Prediction is easier than estimation, at least in the sense of being more forgiving. This allows for the use of inefficient

⁸See Figure 15.5 of Efron and Hastie (2016).

⁹This imagines that we have a single new observation to predict. Suppose instead that we have m new observations $X_1, X_2, \dots, X_m \stackrel{\text{ind}}{\sim} \mathcal{N}(\mu, 1)$, and that we wish to predict their mean $\bar{X}$. With $m = 10$ the efficiency ratio is $E\{(\bar{X} - \check{x})^2\}/E\{(\bar{X} - \bar{x})^2\} = 1.16$; with $m = 100$, 1.46; and with $m = \infty$, 1.57. One can think of estimation as the prediction of future mean values.

estimators like the gbm stumps, that are convenient for mass deployment. The pure prediction algorithms operate nonparametrically, a side benefit of not having to worry much about estimation efficiency.

For the comparison of prediction with attribution we consider an idealized version of a microarray study involving n subjects, $n/2$ healthy controls and $n/2$ sick patients: any one subject provides a vector of measurements on N genes, $\mathbf{X} = (X_1, X_2, \dots, X_N)^t$, with

$$X_j \stackrel{\text{ind}}{\sim} \mathcal{N}(\pm\delta_j/2c, 1) \quad (c = \sqrt{n/4}), \quad (21)$$

for $j = 1, 2, \dots, N$, “plus” for the sick and “minus” for the healthy; δ_j is the effect size for gene j . Most of the genes are null, $\delta_j = 0$, say N_0 of them, but a small number N_1 have δ_j equal a positive value Δ ,

$$N_0 : \delta_j = 0 \quad \text{and} \quad N_1 : \delta_j = \Delta. \quad (22)$$

A new person arrives and produces a microarray of measurements $\mathbf{X} = (X_1, X_2, \dots, X_N)^t$ satisfying (21) but without us knowing the person's healthy/sick status; that is, without knowledge of the $\pm$ value. *Question:* How small can N_1/N_0 get before prediction becomes impossible? The answer, motivated in Appendix A, is that asymptotically as $N_0 \rightarrow \infty$, accurate prediction is possible if

$$N_1 = O(N_0^{1/2}), \quad (23)$$

but not below that.

By contrast, Appendix A shows that effective attribution requires

$$N_1 = O(N_0). \quad (24)$$

In terms of “needles in haystacks” (Johnstone and Silverman 2004), attribution needs an order of magnitude more needles than prediction. The prediction tactic of combining weak learners is not available for attribution, which, almost by definition, is looking for *strong* individual predictors. At least in this example, it seems fair to say that prediction is much easier than attribution.

The three main regression categories can usefully be arranged in order

$$\text{prediction} \quad \cdots \quad \text{estimation} \quad \cdots \quad \text{attribution}, \quad (25)$$

with estimation in a central position and prediction and attribution more remote from each other. Traditionally, estimation is linked to attribution through p -values and confidence intervals, as in Table 1. Looking in the other direction, good estimators, when they are available, are usually good predictors. Both prediction and estimation focus their output on the n side of the $n \times p$ matrix $\mathbf{x}$, while attribution focuses on the p side. Estimation faces both ways in (25).

The randomForest algorithm *does* attempt to connect prediction and attribution. Along with the predictions, an *importance measure*¹⁰ is computed for each of the p predictor

¹⁰There are several such measures. The one in Figure 7 relates to Gini's criterion, Section 3. At the conclusion of the algorithm we have a long list of all the splits in all the bootstrap trees; a single predictor's importance score is the sum of the decreases in the Gini criterion over all splits where that predictor was the splitting variable.

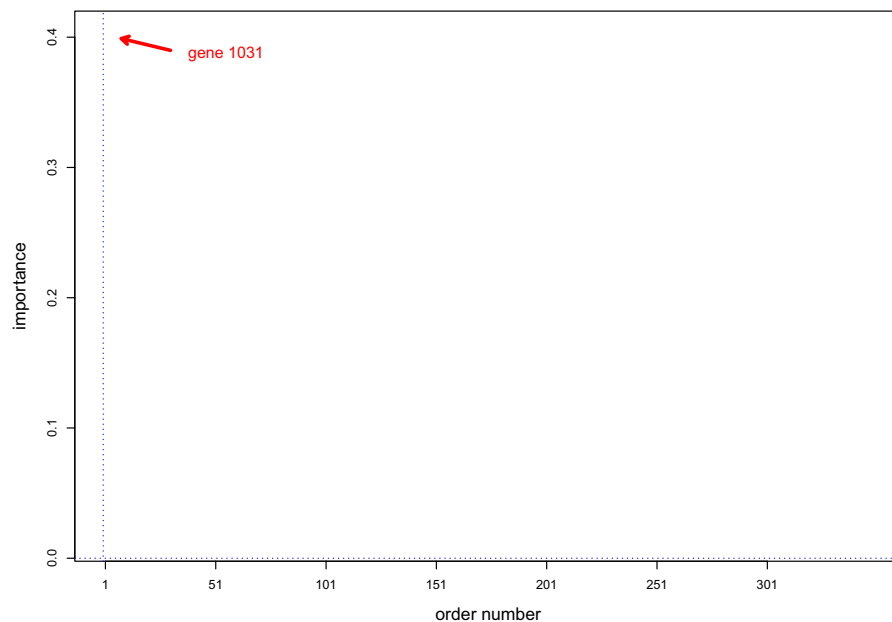


Figure 7. Random forest importance measures for prostate cancer prediction rule of Figure 5, plotted in order of declining importance.

Table 3. Number of test set errors for prostate cancer random forest predictions, removing top predictors shown in Figure 7.

# removed	0	1	5	10	20	40	80	160	348
# errors	1	0	3	1	1	2	2	2	0

variables. Figure 7 shows the ordered importance scores for the prostate cancer application of Figure 5. Of the $p = 6033$ genes, 348 had positive scores, these being the genes that *ever* were chosen as splitting variables. Gene 1031 achieved the most importance, with about 25 others above the sharp bend in the importance curve. Can we use the importance scores for attribution, as with the asterisks in Table 1?

In this case, the answer seems to be no. I removed gene 1031 from the dataset, reducing the data matrix $\mathbf{x}$ to 102×6032 , and reran the `randomForest` prediction algorithm. Now the number of test set prediction errors was zero. Removing the most important five genes, the most important 10, . . . , the most important 348 genes had similarly minor effects on the number of test set prediction errors, as shown in Table 3.

At the final step, *all* of the genes involved in constructing the original prediction rule of Figure 5 had been removed. Now $\mathbf{x}$ was 102×5685 , but the random forest rule based on the reduced dataset $\mathbf{d} = \{\mathbf{x}, \mathbf{y}\}$ still gave excellent predictions. As a matter of fact, there were *zero* test set errors for the realization shown in Table 3. The prediction rule at the final step yielded 364 “important” genes, disjoint from the original 348. Removing all $712 = 348 + 364$ genes from the prediction set—so now $\mathbf{x}$ was 102×5321 —still gave a random forest prediction rule that made only one test set error.

The “weak learners” model of prediction seems dominant in this example. Evidently there are a great many genes weakly correlated with prostate cancer, which can be combined in different combinations to give near-perfect predictions. This is an advantage if prediction is the only goal, but a disadvantage as far as attribution is concerned. Traditional methods of attribution

operate differently, striving as in Table 1 to identify a small set of causal covariates (even if strict causality cannot be inferred).

The pure prediction algorithms’ penchant for coining weakly correlated new predictors moves them in the opposite direction from attribution. Section 9 addresses *sparsity*—a working assumption of there being only a few important predictors—which is not at all the message conveyed by Table 3.

6. The Training/Test Set Paradigm

A crucial ingredient of modern prediction methodology is the training/test set paradigm: the data $\mathbf{d}$ (1) is partitioned into a training set $\mathbf{d}_{\text{train}}$ and a test set $\mathbf{d}_{\text{test}}$; a prediction rule $f(\mathbf{x}, \mathbf{d}_{\text{train}})$ is computed using only the data $\mathbf{d}_{\text{train}}$; finally, $f(\mathbf{x}, \mathbf{d}_{\text{train}})$ is applied to the cases in $\mathbf{d}_{\text{test}}$, yielding an honest estimate of the rule’s error rate. But honest does not mean perfect.

This paradigm was carried out in Section 4 for the prostate cancer microarray study, producing an impressively small error rate estimate of 2% for random forests.¹¹ This seemed extraordinary to me. Why not use this rule to diagnose prostate cancer based on the vector of a new man’s 6033 gene expression measurements? The next example suggests how this might go wrong.

The training and test sets for the prostate cancer data of Section 4 were obtained by *randomly* dividing the 102 men into two sets of 51, each with 25 normal controls and 26 cancer patients. Randomization is emphasized in the literature as a guard against bias. Violating this advice, I repeated the analysis, this time selecting for the training set the 25 normal controls and 26 cancer patients with the lowest ID numbers. The test set was the remaining 51 subjects, those with the highest IDs, and again contained 25 normal controls and 26 cancer patients.

In the reanalysis `randomForest` did not perform nearly as well as in Figure 5: $f(\mathbf{x}, \mathbf{d}_{\text{train}})$ made 12 wrong predictions

¹¹ Taking account of the information in Table 2, a better error rate estimate is 3.7%.

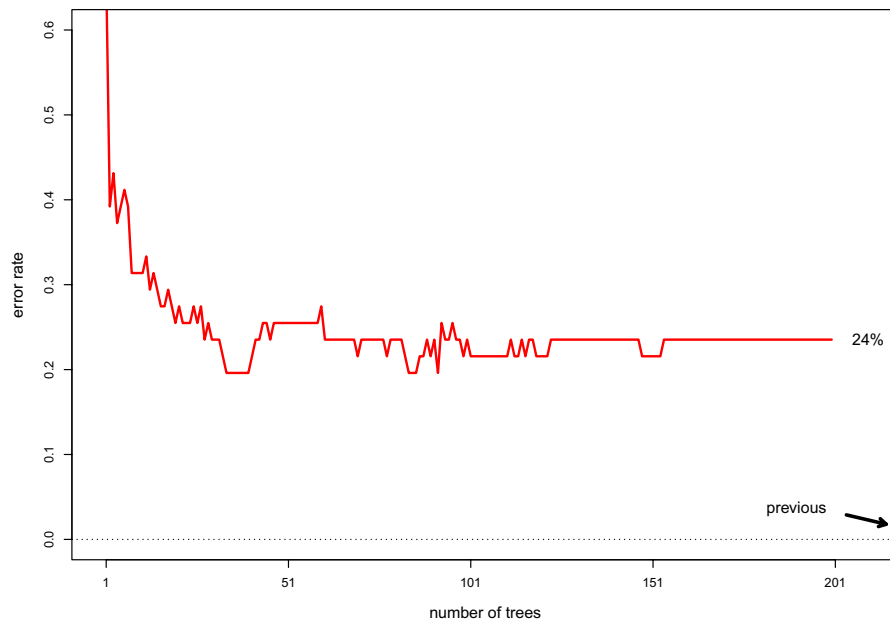


Figure 8. `randomForest` test set error for prostate cancer microarray study, now with training/test sets determined by early/late ID number. Results are much worse than in Figure 5.

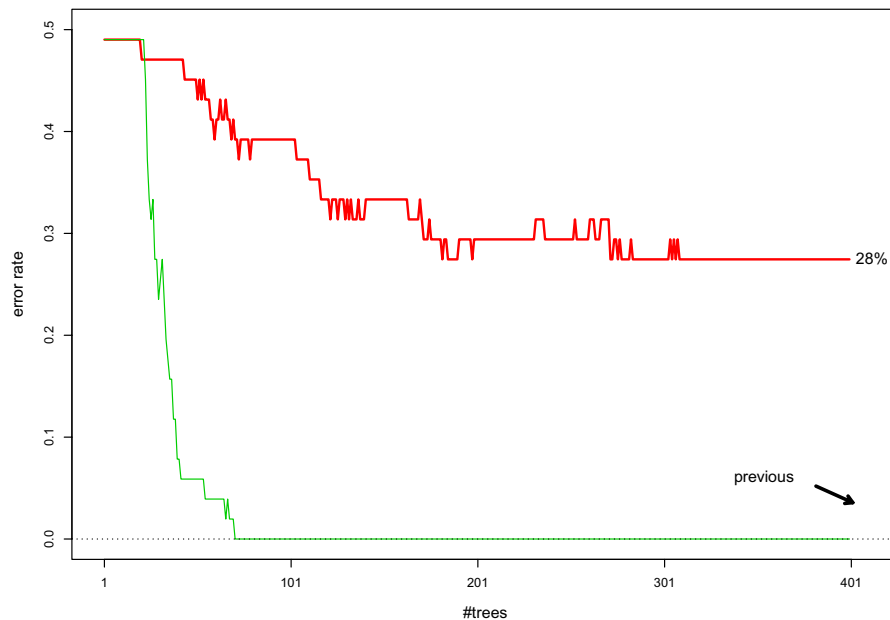


Figure 9. `gbm` test set error, early/late division; compare with Figure 6. Going on to 800 trees decreased error estimate to 26%. Training set error rate, thin curve, was zero after step 70 but test error rate continued to decline. See the brief discussion in Criterion 5 of Section 8.

on $\mathbf{d}_{\text{test}}$ with error rate 24%, rather than the previous 2%, as graphed in Figure 8. The boosting algorithm `gbm` was just as bad, producing prediction error rate 28% (14 wrong predictions) as shown in Figure 9.

Why are the predictions so much worse now? It is not obvious from inspection but the prostate study subjects might have been collected in the order listed,¹² with some small methodological differences creeping in as time progressed. Perhaps all those weak learners going into `randomForest` and `gbm` were

vulnerable to such differences. The prediction literature uses *concept drift* as a label for this kind of trouble, a notorious example being the Google flu predictor, which beat the CDC for a few years before failing spectacularly.¹³ Choosing one's test set by random selection sounds prudent but it is guaranteed to hide any drift effects.

Concept drift gets us into the question of what our various regression methods, new and old, are supposed to be telling us. Science, historically, has been the search for the underlying truths that govern our universe: truths that are supposed to be

¹²A singular value decomposition of the normal-subject data had second principal vector sloping upward with ID number, but this was not true for the cancer patient data.

¹³The CDC itself now sponsors annual Internet-based flu forecasting challenges (Schmidt 2019); see their past results at predict.cdc.gov.

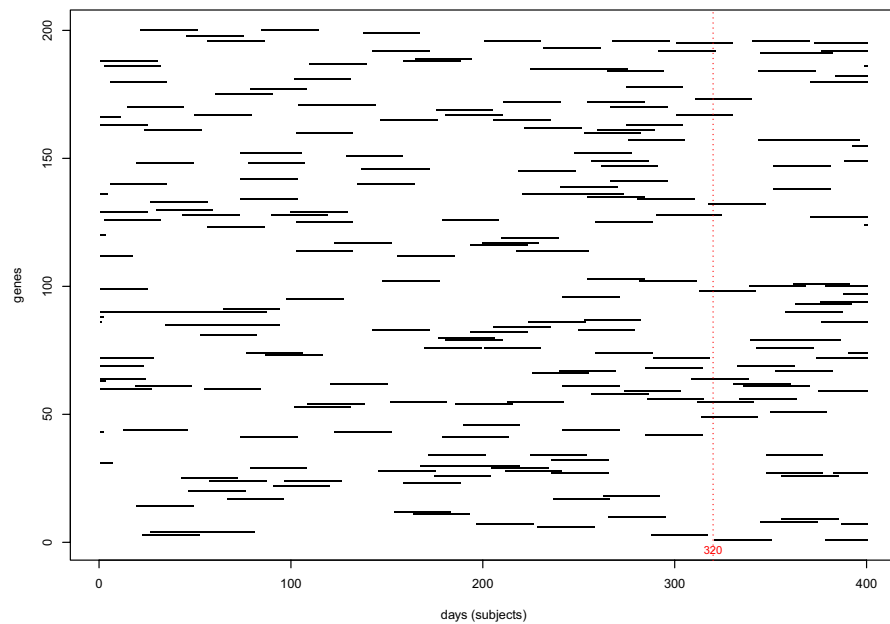


Figure 10. Black line segments indicate active episodes in the hypothetical microarray study. (Matrix transposed for typographical convenience.)

Table 4. Comparing logistic regression coefficients for neonate data for year 1 (as in Table 1) and year 2; correlation coefficient 0.79.

	gest	ap	bwei	resp	cpap	ment	rate	hr	head	gen	temp
Year 1	−0.47	−0.58	−0.49	0.78	0.27	1.10	−0.09	0.01	0.1	0.00	0.02
Year 2	−0.65	−0.27	−0.19	1.13	0.15	0.41	−0.47	−0.02	−0.2	−0.04	0.16

eternal, like Newton's laws. The eternal part is clear enough in physics and astronomy—the speed of light, $E = mc^2$, Hubble's law—and perhaps in medicine and biology, too, for example, DNA and the circulation of blood. But modern science has moved on to fields where truth may be more contingent, such as economics, sociology, and ecology.

Without holding oneself to Newtonian standards, traditional estimation and attribution usually aim for long-lasting results that transcend the immediate datasets. In the surface plus noise paradigm of Section 2, the surface plays the role of truth—at least eternal enough to justify striving for its closest possible estimation.

In the neonate example of Table 1 we would hope that starred predictors like *gest* and *ap* would continue to show up as important in future studies. A second year of data was in fact obtained, but with only $n = 246$ babies. The same logistic regression model was run for the year 2 data and yielded coefficient estimates reasonably similar to the year 1 values; see Table 4. Newton would not be jealous, but something of more than immediate interest seems to have been discovered.

Nothing rules out eternal truth-seeking for the pure prediction algorithms, but they have been most famously applied to more ephemeral phenomena: credit scores, Netflix movie recommendations, facial recognition, *Jeopardy!* competitions. The ability to extract information from large heterogeneous data collections, even if just for short-term use, is a great advantage of the prediction algorithms. Random selection of the test set makes sense in this setting, as long as one does not accept the estimated error rate as applying too far outside the limited range of the current data.

Here is a contrived microarray example where *all* the predictors are ephemeral: $n = 400$ subjects participate in the study, arriving one per day in alternation between Treatment and Control; each subject is measured on a microarray of $p = 200$ genes. The 400×200 data matrix $\mathbf{x}$ has independent normal entries

$$x_{ij} \stackrel{\text{ind}}{\sim} \mathcal{N}(\mu_{ij}, 1) \quad \text{for } i = 1, 2, \dots, 400 \quad \text{and} \\ j = 1, 2, \dots, 200. \quad (26)$$

Most of the μ_{ij} are null, $\mu_{ij} = 0$, but occasionally a gene will have an active episode of 30 days during which

$$\mu_{ij} = 2 \quad \text{for Treatment} \quad \text{and} \quad -2 \quad \text{for Control} \quad (27)$$

for the entire episode, or

$$\mu_{ij} = 2 \quad \text{for Control} \quad \text{and} \quad -2 \quad \text{for Treatment} \quad (28)$$

for the entire episode. The choice between (27) and (28) is random, as is the starting date for each episode. Each gene has expected number of episodes equal 1. The black line segments in Figure 10 indicate all the active time periods.

The 400 hypothetical subjects were *randomly* divided into a training set of 320 and a test set of 80. A randomForest analysis gave the results seen in the left panel of Figure 11, with test set error rate 19%. A second randomForest analysis was carried out, using the subjects from days 1 to 320 for the training set and from days 321 to 400 for the test set. The right panel of Figure 11 now shows test set error about 45%.

In this case, it is easy to see how things go wrong. From any one day's measurements it is possible to predict Treatment or Control from the active episode responses on nearby days

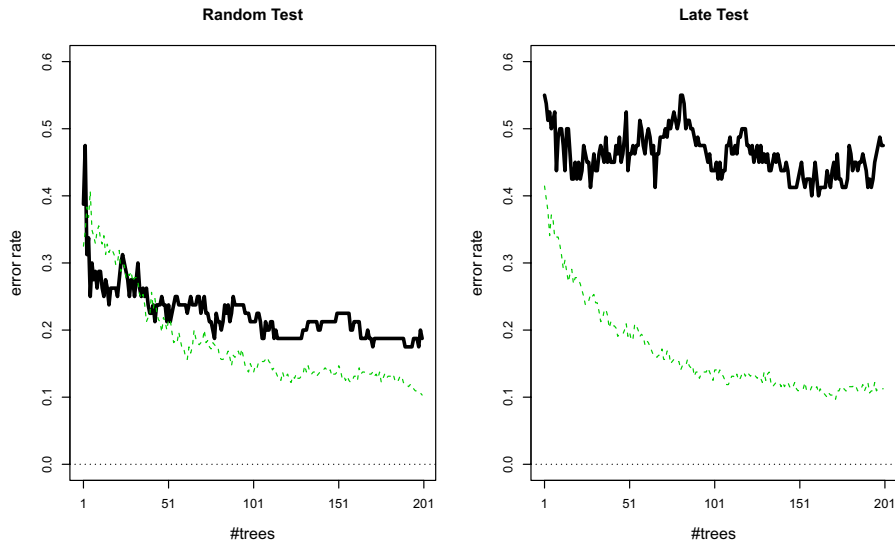


Figure 11. randomForest prediction applied to contrived microarray study pictured in Figure 10. Left panel: Test set of size 80, selected randomly from 400 days; heavy black curve shows final estimated test error rate of 19%. Right panel: Test set days 321–400; now error rate estimate is 45%. Light dotted curves in both panels are training set errors.

(even without knowledge of the activity map in Figure 10). This works for the random training/test division, where most of the test days will be intermixed with training days. Not so for the early/late division, where most of the test days are far removed from training set episodes. To put it another way, prediction is easier for interpolation than extrapolation.¹⁴

What in general can we expect to learn from training/test set error estimates? Going back to formulation (1), the usual assumption is that the pairs (x_i, y_i) are independent and identically distributed (iid) from some probability distribution F on $(p + 1)$ -dimensional space,

$$(x_i, y_i) \stackrel{\text{iid}}{\sim} F \quad \text{for } i = 1, 2, \dots, n. \quad (29)$$

A training set $\mathbf{d}_0$ of size n_0 and a test set $\mathbf{d}_1$ of size $n_1 = n - n_0$ are chosen (*how* is irrelevant under model (29)), rule $f(x, \mathbf{d}_0)$ is computed and applied to $\mathbf{d}_1$, generating an error estimate

$$\widehat{\text{Err}}_{n_0} = \frac{1}{n_1} \sum_{d_1} L(y_i, f(x_i, \mathbf{d}_0)), \quad (30)$$

L some loss function like squared error or counting error. Then, under model (29), $\widehat{\text{Err}}_{n_0}$ is an unbiased estimate of

$$\text{Err}_{n_0}(F) = E_F \{ \widehat{\text{Err}}_{n_0} \}, \quad (31)$$

the average prediction error of a rule¹⁵ $f(x, \mathbf{d}_0)$ formed from n_0 draws from F .

Concept drift can be interpreted as a change in the data-generating mechanism (29), say F changing to some new distribution $\bar{F}$, as seems the likely culprit in the prostate cancer example of Figures 8 and 9.¹⁶ Traditional prediction methods

are also vulnerable to such changes. In the neonate study, the logistic regression rule based on the year 1 data had a cross-validated error rate of 20% which increased to 22% when applied to the year 2 data.

The story is more complicated for the contrived example of Figures 10 and 11, where model (29) does not strictly apply. There the effective predictor variables are ephemeral, blooming and fading over short time periods. A reasonable conjecture (but no more than that) would say the weak learners of the pure prediction algorithms are prone to ephemerality, or at least are more prone than the “main effects” kind of predictors favored in traditional methodology. Whether or not this is true, I feel there is some danger in constructing training/test sets by random selection, and that their error estimates must be taken with a grain of statistical salt. To put things operationally, I’d worry about recommending the random forests prediction rule in Figure 5 to a friend concerned about prostate cancer.

This is more than a hypothetical concern. In their 2019 article, “Deep Neural Networks are Superior to Dermatologists in Melanoma Image Classification,” Brinker et al. demonstrate just what the title says; the authors are justifiably cautious, recommending future studies for validation. Moreover, they acknowledge the limitations of using a randomly selected test set, along with the possible ephemerality of some of the algorithm’s predictor variables. Frequent updating would be necessary for serious use of any such diagnostic algorithm, along with studies to show that certain subpopulations were not being misdiagnosed.¹⁷

7. Smoothness

It was not just a happy coincidence that Newton’s calculus accompanied Newton’s laws of motion. The Newtonian world, as fleshed out by Laplace, is an infinitely smooth one in which

¹⁴Yu and Kumbier (2019) proposed the useful distinction of “internal testing” versus “external testing.”

¹⁵An important point is that “a rule” means one formed according to the algorithm of interest and the data-generating mechanism, not the specific rule $f(x, \mathbf{d}_0)$ at hand; see Figure 12.3 of Efron and Hastie (2016).

¹⁶Cox, in his discussion of Breiman (2001), says of the applicability of model (29): “However, much prediction is not like this. Often the prediction is under quite different ...conditions ...[for example] what would be the effect on annual incidence of cancer in the United States of reducing by 10% the medical use of x-rays? etc.”

¹⁷Facial recognition algorithms have been shown to possess gender, age, and race biases.

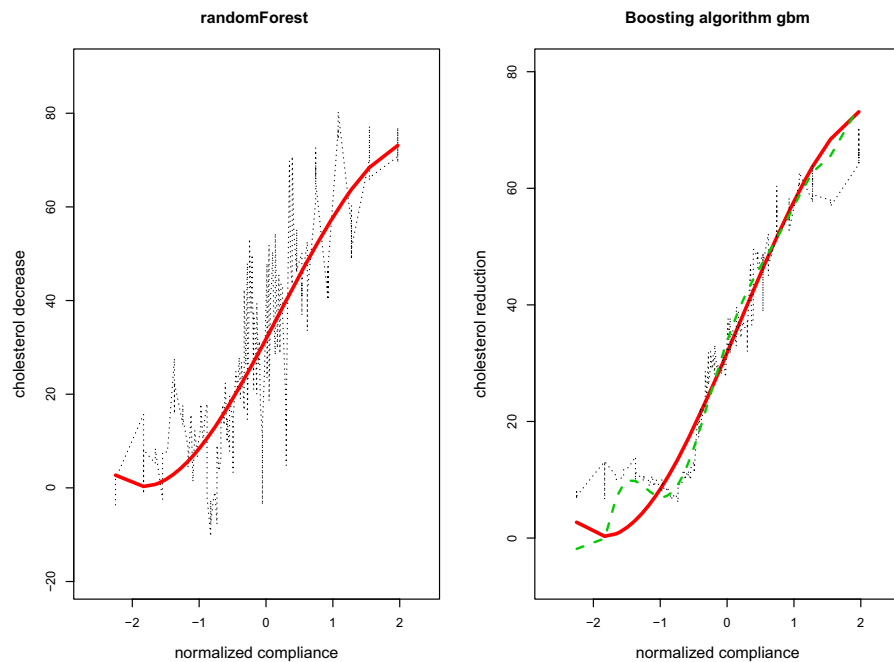


Figure 12. randomForest and gbm fits to the cholestyramine data of Figure 1 in Section 2. Heavy curve is cubic OLS; dashed curve in right panel is 8th degree OLS fit.

small changes in cause yield small changes in effect; a world where derivatives of many orders make physical sense. The parametric models of traditional statistical methodology enforce the smooth-world paradigm. Looking back at Figure 1 in Section 2, we might not agree with the exact shape of the cholestyramine cubic regression curve but the smoothness of the response seems unarguable: going from, say, 1 to 1.01 on the compliance scale should not much change the predicted cholesterol decrease.

Smoothness of response is not built into the pure prediction algorithms. The left panel of Figure 12 shows a randomForest estimate of cholesterol decrease as a function of normalized compliance. It roughly follows the OLS cubic curve but in a jagged, definitely unsmooth fashion. Algorithm gbm, in the right panel, gave a less jagged “curve” but still with substantial local discontinuity.

The choice of *cubic* in Figure 1 was made on the basis of a Cp comparison of polynomial regressions degrees 1 through 8, with cubic best. Both randomForest and gbm in Figure 12 began by taking $\mathbf{x}$ to be the 164×8 matrix `poly(c, 8)` (in R notation), with $\mathbf{c}$ the vector of adjusted compliances—an 8th degree polynomial basis. The light dashed curve in the right panel is the 8th degree polynomial OLS fit, a pleasant surprise being how the gbm predictions follow it over much of the compliance range. Perhaps this is a hopeful harbinger of how prediction algorithms could be used as nonparametric regression estimates, but the problems get harder in higher dimensions.

Consider the *supernova data*: absolute brightness y_i has been recorded for each of $n = 75$ supernovas, as well as x_i a vector of spectral energy measurements at $p = 25$ different wavelengths, so the dataset is

$$\mathbf{d} = \{\mathbf{x}, \mathbf{y}\}, \quad (32)$$

with $\mathbf{x}$ 75×25 and $\mathbf{y}$ a 75-vector. After some preprocessing, a reasonable model is

$$y_i \stackrel{\text{ind}}{\sim} \mathcal{N}(\mu_i, 1). \quad (33)$$

It is desired to predict μ_i from x_i .

Our data $\mathbf{d}$ is unusually favorable in that the 75 supernovas occurred near enough to Earth to allow straightforward determination of y_i without the use of x_i . However, this kind of determination is not usually available, while x_i is always observable; an accurate prediction rule

$$\hat{y}_i = f(x_i, \mathbf{d}) \quad (34)$$

would let astronomers better use Type 1a supernovas as “standard candles” in determining the distances to remote galaxies.¹⁸ In this situation, the smoothness of $f(x, \mathbf{d})$ as a function of x would be a given.

Algorithms randomForest and gbm were fit to the supernova data (32). How smooth or jagged were they? For any two of the 75 cases, say i_1 and i_2 , let $\{x_\alpha\}$ be the straight line connecting x_{i_1} and x_{i_2} in R^{25} ,

$$\{x_\alpha = \alpha x_{i_1} + (1 - \alpha)x_{i_2} \text{ for } \alpha \in [0, 1]\}, \quad (35)$$

and $\{\hat{y}_\alpha\}$ the corresponding predictions. A linear model would yield linear interpolation, $y_\alpha = \alpha y_{i_1} + (1 - \alpha)y_{i_2}$.

Figure 13 graphs $\{y_\alpha\}$ for three cases: $i_1 = 1$ and $i_2 = 3$, $i_1 = 1$ and $i_2 = 39$, and $i_1 = 39$ and $i_2 = 65$. The randomForest traces are notably eccentric, both locally and globally; gbm less so, but still far from smooth.¹⁹

There is no need for model smoothness in situations where the target objects are naturally discrete: movie recommendations, credit scores, chess moves. For scientific applications,

¹⁸The discovery of dark energy and the cosmological expansion of the universe involved treating Type 1a supernovas as always having the same absolute brightness, that is, as being perfect standard candles. This is not exactly true. The purpose of this analysis is to make the candles more standard by regression methods, and so improve the distance measurements underlying cosmic expansion. Efron and Hastie (2016) discussed a subset of this data in their Chapter 12.

¹⁹The relatively smoother results from gbm have to be weighed against the fact that it gave much worse predictions for the supernova data, greatly overshrinking the $\hat{y}_i$ toward zero.

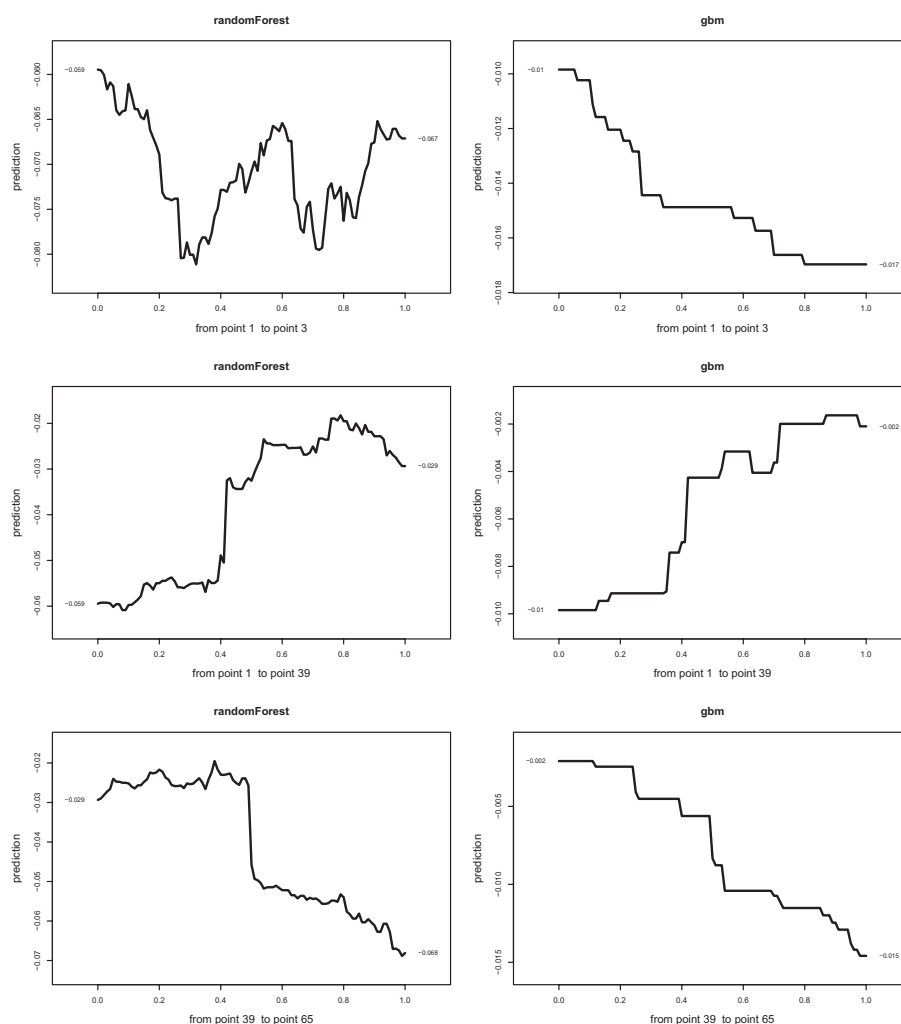


Figure 13. Interpolation between pairs of points in supernova data. Left side is `randomForest`, right side is `gbm`.

at least for some of them, smoothness will be important to a model's plausibility. As far as I know, there is no inherent reason that a pure prediction algorithm must give jagged results. Neural networks, which are essentially elaborate logistic regression programs, might be expected to yield smoother output.

8. A Comparison Checklist

Prediction is not the same as estimation, though the two are often conflated. Much of this article has concerned the differences. As a summary of what has gone before as well as a springboard for broader discussion, this section presents a checklist of important distinctions and what they mean in terms of statistical practice.

The new millennium got off to a strong start on the virtues of prediction with Leo Breiman's (2001) *Statistical Science* publication, "Statistical Modeling: The Two Cultures." An energetic and passionate argument for the "algorithmic culture"—what I have been calling the pure prediction algorithms—in this work Leo excoriated the "data modeling culture" (i.e., traditional methods) as of limited utility in the dawning world of Big Data. Professor David Cox, the lead discussant, countered with a characteristically balanced defense of mainstream statistics, not

rejecting prediction algorithms but pointing out their limitations. I was the second discussant, somewhat skeptical of Leo's claims (which were effusive toward random forests, at that time new) but also somewhat impressed.

Breiman turned out to be more prescient than me: pure prediction algorithms have seized the statistical limelight in the twenty-first century, developing much along the lines Leo suggested. The present paper can be thought of as a continued effort on my part to answer the question of how prediction algorithms relate to traditional regression inference.

Table 5 displays a list of six criteria that distinguish traditional regression methods from the pure prediction algorithms. My previous "broad brush" warning needs to be made again: I am sure that exceptions can be found to all six distinctions, nor are the listed properties written in stone, the only implication being that they reflect current usage.

Criterion 1. Surface plus noise models are ubiquitous in traditional regression methodology, so much so that their absence is disconcerting in the pure prediction world. Neither surface nor noise is required as input to `randomForest`, `gbm`, or their kin. This is an enormous advantage for easy usage. Moreover, you cannot be using a wrong model if there is no model.

Table 5. A comparison checklist of differences between traditional regression methods and pure prediction algorithms.

	Traditional regressions methods	Pure prediction algorithms
1.	Surface plus noise models (continuous, smooth)	Direct prediction (possibly discrete, jagged)
2.	Scientific truth (long-term)	Empirical prediction accuracy (possibly short-term)
3.	Parametric modeling (causality)	Nonparametric (black box)
4.	Parsimonious modeling (researchers choose covariates)	Anti-parsimony (algorithm chooses predictors)
5.	$p \times n$: with $p \ll n$ (homogeneous data)	$p \gg n$, both possibly enormous (mixed data)
6.	Theory of optimal inference (mle, Neyman–Pearson)	Training/test paradigm (Common Task Framework)

NOTE: See commentary in the text.

A clinician dealing with possible prostate cancer cases will certainly be interested in effective prediction, but the disease's etiology will be of greater interest to an investigating scientist, and that's where traditional statistical methods come into their own. If random forests had been around since 1908 and somebody just invented regression model significance testing, the news media might now be heralding an era of "sharp data."

Eliminating surface-building from inference has a raft of downstream consequences, as discussed in what follows. One casualty is smoothness (Section 7). Applications of prediction algorithms have focused, to sensational effect, on discrete target spaces—Amazon recommendations, translation programs, driving directions—where smoothness is irrelevant. The natural desire to use them for scientific investigation may hasten development of smoother, more physically plausible algorithms.

Criterion 2. The two sides of Table 5 use similar fitting criteria—some version of least squares for quantitative responses—but they do so with different paradigms in mind. Following a 200-year-old scientific path, traditional regression methods aim to extract underlying truth from noisy data: perhaps not eternal truth but at least some takeaway message transcending current experience.

Without the need to model surface or noise mechanisms, scientific truth fades in importance on the prediction side of the table. There may not be any underlying truth. Prediction methods can be comfortable with ephemeral relationships that need only remain valid until the next update. To quote Breiman, "The theory in this field shifts focus from data models to the properties of algorithms," that is, from the physical world to the computer. Research in the prediction community, which is an enormous enterprise, is indeed heavily focused on computational properties of algorithms—in particular, how they behave as n and p become huge—and less on how they relate to models of data generation.

Criterion 3. Parametric modeling plays a central role in traditional methods of inference, while the prediction algorithms are nonparametric, as in (29). ("Nonparametric," however, can

involve hosts of *tuning parameters*, millions of them in the case of deep learning, all relating to the algorithm rather than to data generation.) Lurking behind a parametric model is usually some notion of causality. In the cholestyramine example of Figure 1 in Section 2, we are likely to believe that increased ingestion of the drug cholestyramine causes cholesterol to decrease in a sigmoidal fashion, even if strict causality is elusive.²⁰

Abandoning mathematical models comes close to abandoning the historic scientific goal of understanding nature. Breiman states the case bluntly:

Data models are rarely used in this community [the algorithmic culture]. The approach is that nature produces data in a black box whose insides are complex, mysterious, and at least partly unknowable.²¹

The black-box approach has a scientifically anti-intellectual feeling but, on the other hand, scientific understanding may be beside the point if prediction is the only goal. Machine translation offers a useful case study, where there has been a several-decade conflict between approaches based on linguistic analysis of language structure and more-or-less pure prediction methods. Under the umbrella name of statistical machine translation (SMT), this latter approach has swept the field, with Google Translate, for example, currently using a deep learning prediction algorithm.

Traditional statistical education involves a heavy course of probability theory. Probability occupies a smaller portion of the nonparametric pure-prediction viewpoint, with probabilistically simple techniques such as cross-validation and the bootstrap shouldering the methodological burden. Mosteller and Tukey's (1977) book, *Data Analysis and Regression: A Second Course in Statistics*, favored a nonprobabilistic approach to inference that would be congenial to a modern course in machine learning.

Criterion 4. The 11 neonate predictor variables in Table 1 were winnowed down from an initial list of 81, following a familiar path of preliminary testing and discussions with the medical scientists. Parsimonious modeling is a characteristic feature of traditional methodology. It can be crucial for estimation and, especially, for attribution, where it is usually true that the power of discovery decreases as the list of predictors grows.

The pure prediction world is anti-parsimonious. Control of the predictor set, or the "features" as they are called, passes from the statistician to the algorithm, which can coin highly interactive new features such as random forests' tree variables. "The more predictor variables, the more information," said Breiman, an especially accurate forecast of the deep learning era.

I was doubtful. My commentary on Breiman's paper began: "At first glance Leo Breiman's stimulating paper looks like an argument against parsimony and scientific insight, and in favor of black boxes with lots of knobs to twiddle. At second glance it still looks that way, but the paper is stimulating" Impressive

²⁰Efron and Feldman (1991) struggled to make a causality argument, one not accepted uncritically by subsequent authors.

²¹Cox counters: "Formal models are useful and often almost, if not quite, essential for incisive thinking."

results like the `randomForest` and `gbm` predictions for the prostate cancer data, [Figures 5 and 6 in Section 4](#), certainly back up Leo's claim. But it is still possible to have reservations. The coined features seem here to be of the weak learner variety, perhaps inherently more ephemeral than the putative strong learners of [Table 1](#).

This is the suggestion made in [Section 6](#). If the prediction algorithms work by clever combinations of armies of weak learners, then they will be more useful for prediction than estimation or, especially, for attribution (as suggested in [Section 5](#)). "Short-term science" is an oxymoron. The use of prediction algorithms for scientific discovery will depend on demonstrations of their longer-term validity.

Criterion 5. Traditional applications ask that the $n \times p$ data matrix $\mathbf{x}$ (n subjects, p predictors) have n substantially greater than p , perhaps $n > 5 \cdot p$, in what is now called "tall data." The neonate data with $n = 812$ and $p = 12$ (counting the intercept) is on firm ground; less firm is the supernova data of [Section 7](#), with $n = 75$ and $p = 25$. On the other side of [Table 5](#) the pure prediction algorithms allow, and even encourage, "wide data," with $p \gg n$. The prostate cancer microarray study is notably wide, with $n = 102$ and $p = 6033$. Even if we begin with tall data, as with the cholestyramine example, the prediction algorithms widen it by the coining of new features.

How do the prediction algorithms avoid overfitting in a $p \gg n$ situation? There are various answers, none of them completely convincing: first of all, using a test set guarantees an honest assessment of error (but see the discussion of [Criterion 6](#)). Second, most of the algorithms employ cross-validation checks during the training phase. Finally, there is an active research area that purports to show a "self-regularizing" property of the algorithms such that even running one of them long past the point where the training data are perfectly fit, as in [Figure 9 of Section 6](#), will still produce reasonable predictions.²²

Estimation and, particularly, attribution work best with homogeneous datasets, where the (x, y) pairs come from a narrowly defined population. A randomized clinical trial, where the subjects are chosen from a specific disease category, exemplifies strict homogeneity. *Not* requiring homogeneity makes prediction algorithms more widely applicable, and is a virtue in terms of generalizability of results, but a defect for interpretability.

The impressive scalability of pure prediction algorithms, which allows them to produce results even for enormous values of n and p , is a dangerous virtue. It has led to a lust for ever larger training sets. This has a good effect on prediction, making the task more interpolative and less extrapolative (i.e., more like [Figures 5 and 6](#), and less like [Figures 8 and 9](#)) but muddies attempts at attribution.²³

Traditional regression methods take the matrix of predictions $\mathbf{x}$ as a fixed ancillary statistic. This greatly simplifies the theory of parametric regression models; $\mathbf{x}$ is just as random as $\mathbf{y}$ in the pure prediction world, the only probability model being the iid nature of the pairs $(x, y) \sim F$. Theory is more difficult in this world, encouraging the empirical emphasis discussed in [Criterion 6](#). Bayesian statistics is diminished in the a-probabilistic prediction world, leaving a tacit frequentist basis as the theoretical underpinning.

Criterion 6. Traditional statistical practice is based on a century of theoretical development. Maximum likelihood estimation and the Neyman–Pearson lemma are optimality criteria that guide applied methodology. On the prediction side of [Table 5](#), theoretical efficiency is replaced by empirical methods, particularly training/test error estimates.

This has the virtue of dispensing with theoretical modeling, but the lack of a firm theoretical structure has led to "many flowers blooming": the popular pure prediction algorithms are completely different from each other. During the past quarter-century, first neural nets then support vector machines, boosting, random forests, and a reprise of neural nets in their deep learning form have all enjoyed the prediction spotlight. In the absence of theoretical guidance we can probably expect more.

In place of theoretical criteria, various prediction competitions have been used to grade algorithms in the so-called "Common Task Framework." The common tasks revolve around some well-known datasets, that of the Netflix movie recommendation data being best known. None of this is a good substitute for a so-far nonexistent theory of optimal prediction.²⁴

Test sets are an honest vehicle for estimating prediction error, but choosing the test set by random selection from the full set $\mathbf{d}$ (1) may weaken the inference. Even modest amounts of concept drift can considerably increase the actual prediction error, as in the prostate data microarray example of [Section 6](#). In some situations there are alternatives to random selection, for example, by selecting training and test according to early and late collection dates, as in [Figures 8 and 9](#). In the supernova data of [Section 7](#), the goal is to apply a prediction rule to supernovas much farther from Earth, so choosing the more distant cases for the test set could be prudent.

In 1914 the noted astronomer Arthur Eddington,²⁵ an excellent statistician, suggested that mean absolute deviation rather than root mean square would be more efficient for estimating a standard error from normally distributed data. Fisher responded in 1920 by showing that not only was root mean square better than mean absolute deviation, it was better than *any* other possible estimator, this being an early example of his theory of sufficiency.

²²For instance, in an OLS fitting problem with $p > n$ where the usual estimate $\hat{\beta} = (\mathbf{x}^t \mathbf{x})^{-1} \mathbf{x}^t \mathbf{y}$ is not available, the algorithm should converge to the $\hat{\beta}$ that fits the data perfectly, $\mathbf{y} = \mathbf{x} \hat{\beta}$, and has minimum norm $\|\hat{\beta}\|$; see Hastie et al. (2019).

²³An experienced statistician will stop reading an article that begins, "Over one million people were asked..." knowing that a random sample of 1000 would be greatly preferable. This bit of statistical folk wisdom is in danger of being lost in the Big Data era. In an otherwise informative popular book titled, of course, *Big Data*, the authors lose all equilibrium on the question of sample size, advocating for $n = \text{all}$: all the flu cases in the country, all

the books on Amazon.com, all possible dog/cat pictures. "Reaching for a random sample in the age of big data is like clutching at a horsewhip in the era of the motor car." In fairness, the book's examples of $n = \text{all}$ are actually narrowly defined, for example, all the street manholes in Manhattan.

²⁴Bayes' rule offers such a theory, but at a cost in assumptions far outside the limits of the current prediction environment.

²⁵Later famous for his astronomical verification of Einstein's theory of relativity.

Traditional methods are founded on these kinds of parametric insights. The two sides of Table 5 are playing by different rules: the left side functions in a Switzerland of inference, comparatively well ordered and mapped out, while Wild West exuberance thrives on the right. Both sides have much to gain from commerce. Before the 1920s, statisticians did not really understand estimation, and after Fisher's work, we did. We are in the same situation now with the large-scale prediction algorithms: lots of good ideas and excitement, without principled understanding, but progress may be in the air.

9. Traditional Methods in the Wide Data Era

The success of the pure prediction algorithms has had a stimulating effect on traditional theory and practice. The theory, forged in the first half of the twentieth century, was tall-data oriented: small values of n , but even smaller p , often just $p = 1$ or 2. Whether or not one likes prediction algorithms, parts of modern science have moved into the wide-data era. In response, traditional methods have been stretching to fit this new world. Three examples follow.

Big data is not the sole possession of prediction algorithms. Computational genetics can go very big, particularly in the form of a GWAS, genome-wide association study. An impressive example was given by Ikram et al. (2009), in a study concerning the narrowing of blood vessels in the eye.²⁶ The amount of narrowing was measured for $n = 15,358$ individuals; each individual had their genome assessed for about $p = 10^6$ SNPs (single-nucleotide polymorphisms), a typical SNP having a certain choice of ATCG value that occurs in a majority of the population or a minor, less prevalent alternative value. The goal was to find SNPs associated with vascular narrowing.

With $\mathbf{x} = 15,356 \times 10^6$ we are definitely in big data and wide data territory. Surface plus noise models seem out of the question here. Instead, each SNP was considered separately: a linear regression was carried out, with the predictor variable the number of minor polymorphisms in the chromosome pair at that location—0, 1, or 2 for each individual—and response his or her narrowing measure. This gave a p -value p_i against the null hypothesis: *polymorphism at location i has no effect on narrowing*, $i = 1, 2, \dots, 10^6$. The Bonferroni threshold for 0.05 significance is

$$p_i \leq 0.05/10^6. \quad (36)$$

Ikram et al. (2009) displayed their results in a “Manhattan plot” with $z_i = -\log_{10}(p_i)$ graphed against location on the genome. Threshold (36) corresponds to $z_i \geq 7.3$; 179 of the 106 SNPs had $z_i > 7.3$, rejecting the null hypothesis of no effect. These were bunched into five locations on the genome, one of which was borderline insignificant. The authors claimed credit for discovering four novel loci. These might represent four genes implicated in vascular narrowing (though a spike in chromosome 12 is shown to spread over a few adjacent genes).

Instead of performing a traditional attribution analysis with $p = 10^6$ predictors, the GWAS procedure performed 10^6 analyses with $p = 1$ and then used a second layer of inference to interpret the results of the first layer. My next example concerns a more elaborate implementation of the two-layer strategy.

While not 10^6 , the $p = 6033$ features of the prostate cancer microarray study in Section 4 are enough to discourage an overall surface plus noise model. Instead we begin with a separate $p = 1$ analysis for each of the genes, as in the GWAS example. The data (16) for the j th gene is

$$\mathbf{d}_j = \{x_{ij} : i = 1, 2, \dots, 102\}, \quad (37)$$

with $i = 1, 2, \dots, 50$ for the normal controls and $i = 51, 52, \dots, 102$ for the cancer patients.

Under normality assumptions, we can compute statistics z_j comparing patients with controls which satisfy, to a good approximation,²⁷

$$z_j \sim \mathcal{N}(\delta_j, 1), \quad (38)$$

where δ_j is the *effect size* for gene j : δ_j equals 0 for “null genes,” genes that show the same genetic activity in patients and controls, while $|\delta_j|$ is large for the kinds of genes being sought, namely, those having much different responses for patients versus controls.

Inferences for the individual genes by themselves are immediate. For instance,

$$p_j = 2\Phi(-z_j) \quad (39)$$

is the two-sided p -value for testing $\delta_j = 0$. However, this ignores having 6033 p -values to interpret simultaneously. As with the GWAS, a second layer of inference is needed.

A Bayesian analysis would hypothesize a prior “density” $g(\delta)$ for the effect size, where g includes an atom of probability π_0 at $\delta = 0$ to account for the null genes. Probably, most of the genes have nothing to do with prostate cancer so π_0 is assumed to be near 1. The *local false discovery rate* $\text{fdr}(z)$ —that is, the probability of a gene being null given z -value z —is, according to Bayes rule,

$$\text{fdr}(z) = \pi_0 \phi(z - \delta) / f(z) \doteq \phi(z - \delta) / f(z), \quad (40)$$

where $\phi(z) = \exp\{-z^2/2\}/\sqrt{2\pi}$, and $f(z)$ is the marginal density of z ,

$$f(z) = \int_{-\infty}^{\infty} \phi(z - \delta) g(\delta) d\delta. \quad (41)$$

The prior $g(\delta)$ is most often unknown. An *empirical Bayes* analysis supposes $f(z)$ to be in some parametric family $f_{\hat{\beta}}(z)$; the MLE $\hat{\beta}$ is obtained by fitting $f_{\hat{\beta}}(\cdot)$ to $\{z_1, z_2, \dots, z_p\}$, the observed collection of all 6033 z -values, giving an estimated false discovery rate

$$\widehat{\text{fdr}}(z) = \phi(z - \delta) / f_{\hat{\beta}}(z). \quad (42)$$

²⁶Microvascular narrowing is thought to contribute to heart attacks, but it is difficult to observe in the heart; observation is much easier in the eye.

²⁷If t_j is the two-sample t -statistic comparing patients with controls, we take $z_j = \Phi^{-1}F_{100}(t_j)$, where F_{100} is the cdf of a t -statistic with 100 degrees of freedom and Φ is the standard normal cdf. Effect size δ_j is a monotone function of the difference in expectations between patients and controls; see Section 7.4 of Efron (2010).

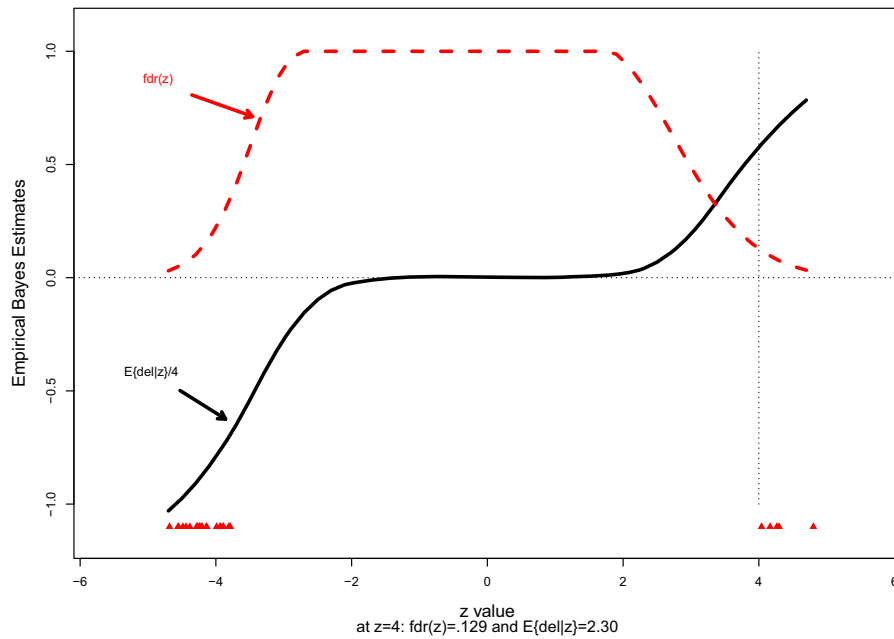


Figure 14. Estimated local false discovery curve $\widehat{fdr}(z)$ and posterior effect size estimate $\widehat{E}\{\delta|z\}$ from empirical Bayes analysis of prostate cancer data (the latter divided by 4 for display purposes). Triangles indicate 29 genes having $\widehat{fdr}(z) \leq 0.20$; circles are 29 most significant genes from `glmnet` analysis.

The dashed curve in Figure 14 shows $\widehat{fdr}(z)$ based on a fifth-degree log-polynomial model for f_β ,

$$\log\{f_\beta(z)\} = \beta_0 + \sum_{k=1}^5 \beta_k z^k; \quad (43)$$

$\widehat{fdr}(z)$ is seen to be near 1.0 for $|z| \leq 2$ (i.e., gene almost certainly null) and declines to zero as $|z|$ grows large, for instance, equaling 0.129 at $z = 4$. The conventional threshold for attributing significance is $\widehat{fdr}(z) \leq 0.20$; 29 genes achieved this, as indicated by the triangles in Figure 14.

We can also estimate the expected effect size. *Tweedie's formula* (Efron 2011) gives a simple expression for the posterior expectation,

$$E\{\delta|z\} = z + \frac{d}{dz} \log f(z), \quad (44)$$

$f(z)$ the marginal density (41). Substituting $f_{\widehat{\beta}}$ for f gave the estimate $E\{\delta|z\}$ in Figure 14. It is nearly zero for $|z| \leq 2$, rising to 2.30 at $z = 4$.

By using a two-level hierarchical model, the empirical Bayes analysis reduces the situation from $p = 6033$ to $p = 5$. We are back in the comfort zone for traditional methods, where parametric modeling for estimation and attribution works well. Both are illustrated in Figure 14.

Sparsity offers another approach to wide-data estimation and attribution: we assume that most of the p predictor variables have no effect and concentrate effort on finding the few important ones. The *lasso* (Tibshirani 1996) provides a key methodology. In an OLS type problem we estimate β , the p -vector of regression coefficients, by minimizing

$$\frac{1}{n} \sum_{i=1}^n (y_i - x_i^t \beta)^2 + \lambda \|\beta\|_1, \quad (45)$$

where $\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$.

Here λ is a fixed tuning parameter: $\lambda = 0$ corresponds to the ordinary least squares solution for β (if $p \leq n$) while $\lambda = \infty$ makes $\widehat{\beta} = 0$. For large values of λ , only a few of the coordinates $\widehat{\beta}_j$ will be nonzero. The algorithm begins at $\lambda = \infty$ and decreases λ , admitting one new nonzero coordinate $\widehat{\beta}_j$ at a time. This works even if $p > n$.

The lasso was applied to the supernova data of Section 7, where $\mathbf{x}$ has $n = 75$ and $p = 25$. Figure 15 shows the first six steps, tracking the nonzero coefficients $\widehat{\beta}_j$ as new variables were added. Predictor 15 was selected first, then 16, 18, 22, 8, and 6, going on to the full OLS solution $\widehat{\beta}$ at step 25. An accuracy formula suggested step 4, with the only nonzero coefficients 15, 16, 18, and 22, as giving the best fit. (These correspond to energy measurements in the iron portion of the spectrum.)

Sparsity and the lasso take us in a direction opposite to the pure prediction algorithms. Rather than combining a myriad of weak predictors, inference is based on a few of the strongest explanatory variables. This is well suited to attribution but less so for prediction.

An R program for the lasso, `glmnet`, was applied to the prostate cancer prediction problem of Section 4, using the same training/test split as that for Figure 5. It performed much worse than `randomForest`, making 13 errors on the test set. Applied to the entire dataset of 102 men, however, `glmnet` gave useful indications of important genes: the circles in Figure 14 show z -values for the 29 genes it ranked as most influential. These have large values of $|z|$, even though the algorithm did not “know” ahead of time to take t -statistics between the cancer and control groups.

The lasso produced *biased* estimates of β , with the coordinate values $\widehat{\beta}_j$ shrunk toward zero. The criticism leveled at prediction methods also applies here: biased estimation is not yet on a firm theoretical footing.

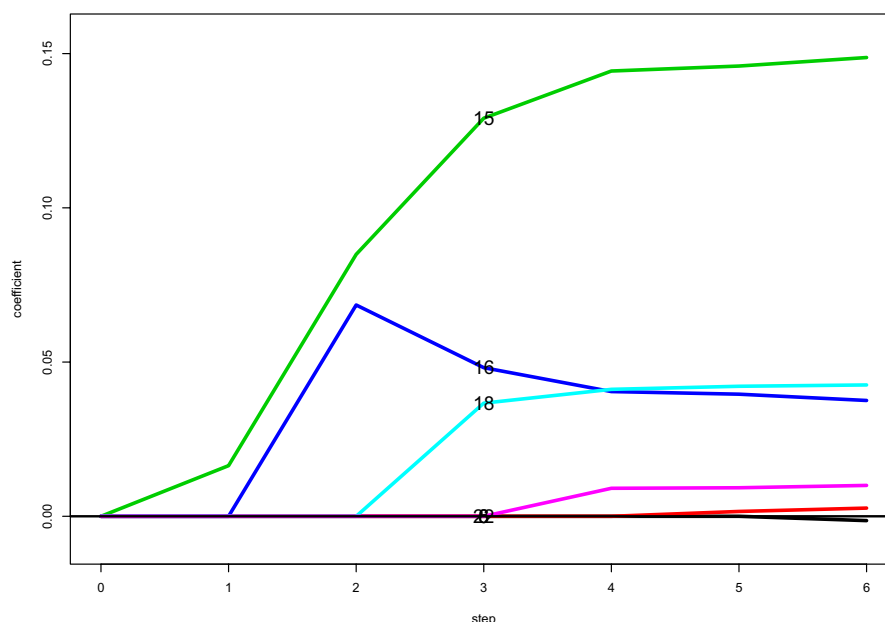


Figure 15. First 6 steps of the lasso algorithm applied to the supernova data; coefficients of various predictors are plotted as a function of step size. Predictor 15 was chosen first, followed by 16, 18, 22, 8, and 6. Stopping at step 4 gives the lowest Cp estimate of estimation error.

10. Two Hopeful Trends

This was not meant to be an “emperor has no clothes” kind of story, rather “the emperor has nice clothes but they’re not suitable for every occasion.” Where they *are* suitable, the pure prediction algorithms can be stunningly successful. When one reads an enthusiastic AI-related story in the press, there’s usually one of these algorithms, operating in enormous scale, doing the heavy lifting. Regression methods have come a long and big way since the time of Gauss.

Much of this article has been concerned with what the prediction algorithms *cannot* do, at least not in their present formulations. Their complex “black box” nature makes the algorithms difficult to critique. Here I have tried to use relatively small datasets (by prediction literature standards) to illustrate their differences from traditional methods of estimation and attribution. The criticisms, most of which will not come as a surprise to the prediction community, were summarized in the six criteria of Table 5 in Section 8.

Some time around the year 2000 a split opened up in the world of statistics.²⁸ For the discussion here we can call the two branches “pure prediction” and “GWAS”: both accommodating huge datasets, but with the former having become fully algorithmic while the latter stayed on a more traditional math-modeling path. The “two hopeful trends” in the title of this section refer to attempts at reunification, admittedly not yet very far along.

Trend 1 aims to make the output of a prediction algorithm more interpretable, that is, more like the output of traditional statistical methods. Interpretable surfaces, particularly those of linear models, serve as the ideal for this achievement. Something like attribution is also desired, though usually not in the specific sense of statistical significance.

One tactic is to use traditional methods for the analysis of a prediction algorithm’s output; see Hara and Hayashi (2016) and Efron and Hastie (2016, p. 346). Wager, Hastie, and Efron (2014) used bootstrap and jackknife ideas to develop standard error calculations for random forest predictions. Murdoch et al. (2019) and Vellido, Martín-Guerrero, and Lisboa (2012) provided overviews of interpretability, though neither focused on pure prediction algorithms. Using information theoretic ideas, Achille and Soatto (2018) discussed statistical sufficiency measures for prediction algorithms.

Going in the other direction, Trend 2 moves from left to right in Table 5, hoping to achieve at least some of the advantages of prediction algorithms within a traditional framework. An obvious target is scalability. Qian et al. (2019) provided a `glmnet` example with $n = 500,000$ and $p = 800,000$. Hastie, Tibshirani, and Friedman (2009) successfully connected boosting to logistic regression. The traditional parametric model that has most currency in the prediction world is logistic regression, so it is reasonable to hope for reunification progress in that area.

“Aspirational” might be a more accurate word than “hopeful” for this section’s title. The gulf seen in Table 5 is wide and the reunification project, if going at all, is just underway. In my opinion, the impediments are theoretical ones. Maximum likelihood theory provides a lower bound on the accuracy of an estimate, and a practical way of nearly achieving it. What can we say about prediction? The Common Task Framework often shows just small differences in error rates among the contestants, but with no way of knowing whether some other algorithm might do much better. In short, we do not have an optimality theory for prediction.

The talks I hear these days, both in statistics and biostatistics, bristle with energy and interest in prediction algorithms. Much of the current algorithmic development has come from outside the statistics discipline but I believe that future progress, especially in scientific applicability, will depend heavily on us.

²⁸See the triangle diagram in the epilogue of Efron and Hastie (2016).

Appendix A

A.1. Out-of-Bag (OOB) Estimates, Section 3

A random forest application involves B nonparametric bootstrap samples from the original dataset $\mathbf{d}$ (1), say $\mathbf{d}^{*1}, \mathbf{d}^{*2}, \dots, \mathbf{d}^{*B}$. Let

$$\hat{y}_{ik} = f(x_i, \mathbf{d}^{*k}), \quad i = 1, 2, \dots, n \quad \text{and} \quad k = 1, 2, \dots, B, \quad (\text{A.1})$$

be the estimate for the i th case obtained from the prediction rule based on the k th bootstrap sample. Also, let

$$N_{ik} = \text{number of times case } (x_i, y_i) \text{ appears in } \mathbf{d}^{*k}. \quad (\text{A.2})$$

The OOB estimate of prediction errors is

$$\widehat{\text{Err}}_0 = \sum_{k=1}^B \sum_{i: N_{ik}=0} L(y_i, \hat{y}_{ik}) / \sum_{k=1}^B \sum_{i: N_{ik}=0} 1, \quad (\text{A.3})$$

where $L(y_i, \hat{y}_{ik})$ is the loss function; that is, $\widehat{\text{Err}}_0$ is the average loss over all cases in all bootstrap samples where the sample did not include the case.

“Bagging” stands for “bootstrap aggregation,” the averaging of the true predictors employed in random forests. The intuition behind (A.3) is that cases having $N_{ik} = 0$ are “out of the bag” of $\mathbf{d}^{*k}$, and so form a natural cross-validation set for error estimation.

A.2. Prediction Is Easier Than Attribution

We wish to motivate (23)–(24), beginning with model (21)–(22). Let $\bar{x}_{0j}$ be the average of the $n/2$ healthy control measurements for gene j , and likewise $\bar{x}_{1j}$ for the $n/2$ sick patient measurements. Then we can compute z values

$$z_j = c(\bar{x}_{1j} - \bar{x}_{0j}) \stackrel{\text{ind}}{\sim} \mathcal{N}(\delta_j, 1) \quad (\text{A.4})$$

for $j = 1, 2, \dots, N$. Letting $\pi_0 = N_0/N$ and $\pi_1 = N_1/N$ be the proportions of null and nonnull genes, (22) says that the z_j take on two possible densities, in proportions π_0 and π_1 ,

$$\pi_0 : z \sim f_0(z) \quad \text{or} \quad \pi_1 : z \sim f_1(z), \quad (\text{A.5})$$

where $f_0(z)$ is the standard normal density $f(z)$, and $f_1(z) = \phi(z - \Delta)$. With both N_0 and N_1 going to infinity, we can and will think of π_0 and π_1 as prior probabilities for null and nonnull genes.

The posterior moments of δ_j given z_j , obtained by applying Bayes rule to model (A.4), have simple expressions in terms of the log derivatives of the marginal density

$$f(z) = \pi_0 f_0(z) + \pi_1 f_1(z), \quad (\text{A.6})$$

$$d(z) \equiv E\{\delta|z\} = z + \frac{d}{dz} \log f(z),$$

and

$$v(z) \equiv \text{var}\{\delta|z\} = 1 + \frac{d^2}{dz^2} \log f(z). \quad (\text{A.7})$$

See Efron (2011). That is,

$$\delta_j|z_j \sim (d_j, v_j), \quad (\text{A.8})$$

where the parenthetical notation indicates the mean and variance of a random quantity and $d_j = d(z_j)$, $v_j = v(z_j)$. Then $X_j \sim \mathcal{N}(\delta_j/2c, 1)$ for the sick subjects in (21) gives

$$X_j|z_j \stackrel{\text{ind}}{\sim} \left(\frac{d_j}{2c}, \frac{v_j}{4c^2} + 1 \right). \quad (\text{A.9})$$

(The calculations which follow continue to focus on the “plus” case of (21).)

A linear combination

$$S = \sum_{j=1}^N w_j X_j \quad (\text{A.10})$$

has posterior mean and variance

$$S \sim \left(\sum_{j=1}^N w_0 A_0, \sum_{j=1}^N w_j^2 B_j \right), \quad (\text{A.11})$$

with

$$A_j = \frac{d_j}{2c} \quad \text{and} \quad B_j = \frac{v_j}{4c^2} + 1.$$

For the “minus” arm of (21) (the healthy controls), $S \sim (-\sum_{j=1}^N w_j A_j, \sum_{j=1}^N w_j^2 B_j)$. We can use S as a prediction statistic, predicting sick for $S > 0$ and healthy for $S < 0$.

The probability of a correct prediction depends on

$$R^2 = \text{mean}(S)^2 / \text{var}(S) = \left(\sum_{j=1}^N w_j A_j \right)^2 / \left(\sum_{j=1}^N w_j^2 B_j \right). \quad (\text{A.12})$$

This is maximized for $w_j = A_j/B_j$ (as with Fisher’s linear discriminant function), yielding

$$S \sim \left(\sum_{j=1}^N A_j^2/B_j, \sum_{j=1}^N A_j^2/B_j \right) \quad \text{and} \quad R^2 = \sum_{j=1}^N A_j^2/B_j. \quad (\text{A.13})$$

A normal approximation for the distribution of S gives

$$\Phi(-R) \quad (\text{A.14})$$

as the approximate probability of a prediction error of either kind; see Efron (2009).

It remains to calculate R^2 .

Lemma 1. A continuous approximation to the sum $R^2 = \sum_{j=1}^N A_j^2/B_j$ is

$$R^2 = \frac{N_1^2}{N_0} \int_{-\infty}^{\infty} \frac{\Delta^2}{4c^2} \frac{N_0/N_1}{1 + \frac{N_0 f_0(z)}{N_1 f_1(z)}} \frac{1}{1 + \frac{v(z)}{4c^2}} f_1(z) dz. \quad (\text{A.15})$$

Before verifying the lemma, we note that it implies (23): letting $N_0 \rightarrow \infty$ with $N_1 = O(N_0)$, and using (A.4), gives

$$R^2 = \frac{N_1^2}{N_0} \int_{-\infty}^{\infty} \frac{\Delta^2}{4c^2} \frac{\exp\{-(z^2/2) + 2\Delta z - (\Delta^2/2)\}}{\sqrt{2\pi} (1 + v(z)/4c^2)} dz. \quad (\text{A.16})$$

The variance $v(z)$ is a bounded quantity under (A.4)—it equals 0.25 for $\Delta = 1$, for instance—so the integral is a finite positive number, say $I(\Delta)$. If $N_1 = \gamma N_0^{1/2}$, then the prediction error probabilities are approximately $\Phi(-\gamma I(\Delta)^{1/2})$ according to (A.14). However, $N_1 = o(N_0^{1/2})$ gives $R^2 \rightarrow 0$ and error probabilities $\rightarrow 1/2$.

It remains to verify the lemma. Since any δ_j equals either 0 or Δ (22),

$$\begin{aligned} d(z) &= E\{\delta|z\} = \Pr\{\delta \neq 0 | z\} \Delta \\ &= \text{tdr}(z) \Delta, \end{aligned} \quad (\text{A.17})$$

where $\text{tdr}(z)$ is the *true discovery rate* $\Pr\{\delta \neq 0 | z\}$,

$$\begin{aligned} \text{tdr}(z) &= \frac{\pi_1 f_1(z)}{\pi_0 f_0(z) + \pi_1 f_1(z)} = \frac{1}{1 + \frac{\pi_0 f_0(z)}{\pi_1 f_1(z)}} \\ &= \frac{1}{1 + \frac{N_0 f_0(z)}{N_1 f_1(z)}}. \end{aligned} \quad (\text{A.18})$$

From (A.11) and (A.13), we get

$$\begin{aligned} R^2 &= \sum_1^N \frac{A_j^2}{B_j} = \sum_1^N \frac{d_j^2}{4c^2 + v_j} \\ &= \sum_1^N \frac{\text{tdr}_j^2}{4c^2 + v_j} = N \int_{-\infty}^{\infty} \frac{\text{tdr}(z)^2 \Delta^2}{4c^2 + v(z)} f(z) dz, \end{aligned} \quad (\text{A.19})$$

the last equality being the asymptotic limit of the discrete sum.

Since $\text{tdr}(z)f(z) = \pi_1 f_1(z) = (N_1/N_0)f_1(z)$, (A.18) becomes

$$R^2 = N_1 \int_{-\infty}^{\infty} \frac{\text{tdr}(z) \Delta^2}{4c^2 + v(z)} f_1(z) dz. \quad (\text{A.20})$$

Then (A.17) gives the lemma. Moreover, unless $N_1 = O(N_0)$, (A.17) shows that $\text{tdr}(z) \rightarrow 0$ for all z , supporting (24).

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Discussion of the Paper “Prediction, Estimation, and Attribution” by B. Efron

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We enjoyed reading Professor Efron’s (Brad) paper just as much as we enjoyed listening to his June 2019 lecture in Leiden. One of the core values underlying statistical research is in how it enables scientific understanding and discoveries, and we appreciate that Efron has placed “science” at the center of his article. Or at least, the discoveries of associations that are not just valid here and now—in the particular dataset I have at hand—but are likely to be stable and replicated under slightly different circumstances.

The successes of purely predictive algorithms (gradient boosting, deep learning) in object recognition, speech recognition, machine translation, and others, are real.¹ Incidentally, when one examines these successes as reported in the media, one observes that they all have in common a nearly infinite sample size (consider that WhatsApp record billions of conversations every year) and a nearly infinite signal-to-noise ratio (SNR) or, put differently, very little uncertainty. If you give me a photograph, there is little debate as to whether there is a cow on it or not. These instances are far from the settings statisticians are accustomed to: moderate sample sizes, a higher degree of uncertainty, and variables/factors which have not been measured—prosaically, things we cannot see. Using Efron’s example, it is not because I know how much you comply with a treatment that I can be certain about how much it will be of benefit to you.

Interestingly, Efron puts the pure predictive algorithms to the test in a scenario where the sample size is extremely moderate by today’s standards ($n = 102$). In the prostate cancer microarray study, he observes the following: split the sample at random as to create a training and test set with the same number of cases and controls, and train a random forest. Perhaps to his surprise, random forests perform extremely well correctly classifying all 51 test patients but one! Surely, we must have “learned” something valuable. Or have we? His second finding is this: if we split patients according to what we believe is the time of entry in the study, then the performance is spectacularly degraded; that is, he now records 12 wrong predictions corresponding to an error rate of about one in four. So what have we really learned? This is all the more discomfiting since prediction rules are intended to be deployed in the second scenario, and not in the highly idealized setting where test and training samples are exchangeable or, said differently, where current data are representative of future data. As far as

we are concerned, this is the key take-way message from his article.

Efron is right to point out that in case we observe good performance on a random train/test split, we should probably not cork champagne. He introduces a healthy dose of reality and we applaud him for this. In high-stakes applications, we should systematically and empirically tests prediction rules and evaluate their robustness just as science tests theories and hypotheses. This of course applies irrespective of the nature of the learning algorithm, for example, whether it is parametric or not. In Efron’s example, there must be a clear shift in the distribution of the data and it is important to understand it if we had the intention of using the learned model in real life. We regard the modeling of distributional shifts and the performance monitoring of predictive models as one of the most important activities statisticians need to engage in. How do we evaluate algorithms beyond the simple metric of checking a classification error on a test set? In our view, algorithmic developments need perhaps to focus less on a single objective and more on appropriate metrics of stability/robustness and reproducibility.

We wrote that Efron put “science” at the center of his article. To maximize the resonance of this choice, and articulate the thoughts that the article provoked in us, we find it useful to recall what Tukey (1962) identified as three constituents of science: “(1) intellectual content, (2) organization into an understandable form, (3) reliance upon the test of experience as the ultimate standard of validity.”

Efron contrasts model-less predictive rules with parametric models, the latter offering, in his view, several advantages, that have to do with all the three constituents of science. Certainly the surface plus noise view of the world—where the surface is described by a sparse parametric model on whose estimation we are to focus—represents a “more understandable form” than the black-boxes at the core of many predictive rules. There are however limits to simplicity. The phenomena contemporary science studies are ever more complex and our data gathering is more comprehensive than ever before. To give an example, we have by now (practically) found all the genes behind Mendelian monogenic diseases and we are now focusing on complex diseases, which are going to be influenced by many genes, possibly interacting. Hence, the phenomena under study may be far too rich to be described with a few parameters. If this is the case, why would we want to estimate parameters that only

exist in the analyst's mind? Similarly, if the model is incorrect, precise inference about these parameters is of dubious scientific merit.

In this context, it is easy to understand the allure of “prediction” as a tool to assess the validity of any strategy to learn from data: the focus is on observable variables, rather than on the estimation of some parameters that might not actually correspond to any existing entities. The importance of prediction in science is, of course, nothing new. In the scientific method, theories are tested by deriving predictions and cross checking with empirical data. The history of statistical reasoning itself documents an attention to prediction. An example is de Finetti's concept of exchangeable random variables, which emphasizes that the realizations of these random variables are all that we ever get to observe. In contrast, a model for how they are generated is by and large a construct of our imagination, and the rational validity of any mental model is predicated on the coherence of the rules with which we rely on past observations to predict the value of future ones (de Finetti 1992). More recently, and with a different emphasis, a related point is made by the authors of “veridical data science,” who identify predictability as a core principle of data science (Yu and Kumbier 2020).

Without contradicting any of this, Efron points out that simply comparing the predictive performance of different strategies might not be as fruitful as one might think. It is hard to know when a predictive model works because everything is drowned in the incompressible randomness of new observations. Often, the difference between competing algorithms can be seen only on the second digits of misclassification error (expressed in percentage so the fourth digit), even if they encode substantially different pictures of reality. How can one learn from this?

We believe that the problem is not with “prediction” per se, but on how we measure performance: one has to take Tukey's call for “understandable form” and “test of experience” more seriously. For prediction to provide actionable information, we need to be able to answer questions as: how accurate are the results of a prediction model? Does the margin of error changes with the values of some predictors? There is much work to be done both in developing *effective ways of measuring and communicating errors*, and in designing algorithms that lead to robust performance. On the first front, resampling-based techniques (the bootstrap (Efron 1979)) and techniques based only upon an exchangeability assumption (the jackknife (Quenouille 1949, 1956), conformal inference (Vovk, Gammerman, and Saunders 1999; Papadopoulos et al. 2002; Lei and Wasserman 2014; Lei et al. 2018)) have already proven powerful. By now we have concrete and efficient methods for constructing predictive intervals $\hat{C}(x)$ from a training sample such that for a new test point $(X_{\text{test}}, Y_{\text{test}})$, we have

$$\mathbb{P}(Y_{\text{test}} \in \hat{C}(X_{\text{test}}) \mid X_{\text{test}} \in \mathcal{X}) \geq 1 - \alpha,$$

where the probability is averaged over both the training sample and the test point. (The statement above holds with the proviso that $\mathcal{X}$ is not too complicated as in the case where $\mathcal{X}$ is the whole sample space, see Barber et al. (2019a).) In short, we can use model-less learning algorithms (black boxes) to predict with confidence. This is an active area of research (Barber et al. 2019b;

Romano, Patterson, and Candès 2019) and further advances can be expected. The careful reader might tick at the notion of exchangeable samples per our earlier discussion. We note that researchers take this issue seriously as well, see Tibshirani et al. (2019) which adapts the ideas of conformal inference to work under covariate shifts. In the latter reference, parametric models are found to be useful to capture distributional shifts.

Efron uses the term “*attribution*” to describe the goal that is often pursued with tests of hypotheses. We like the word choice and we agree that attribution is important. Being able to identify at least the variables that are stably associated with an outcome both increases our understanding and offers a precise way to adopt the test of experience as the ultimate standard of validity. There is not much utility in capturing “ephemeral” associations. The “replicability crisis” which has received so much attention in both the technical and popular media, makes it abundantly clear. The good news is that one can perform attribution without relying on the truth of a parametric models. In the knockoff framework, one can use *any* black-box (any predictive model) to identify any kind possible association—linear or not—between a large set of explanatory variables $X = (X_1, X_2, \dots, X_p)$ and an outcome Y . This framework rigorously tests whether the conditional distribution $Y \mid X$ actually depends on any of the variables X_j without assuming any model for how X and Y are connected (Candès et al. 2018). The burden of knowledge is here on the distribution of the covariates and there are many problems for which good models are at hand. This is the case in genetic studies where large collections of genotype data are available and we have models of linkage disequilibrium to describe their distribution (Sesia, Sabatti, and Candès 2019; Sesia et al. 2020). In such studies, conditional testing—as opposed to the marginal testing discussed in Efron's paper—seems far closer to the scientific question of interest, see Bates et al. (2020) for links with causality.

Returning to Tukey's point about intellectual content, perhaps the most powerful form of knowledge is the ability to act on a system to modify its outcome. This requires having a handle on causal effects. Recent years have witnessed an explosion of research on causal inference methods, many of which try to leverage the power of machine learning algorithms (Wager and Athey 2018; Rothenhäusler, Bühlmann, and Meinshausen 2019).

Efron offered a useful and powerful reminder of what are the ultimate goals of data analysis, and what increased knowledge looks like (or does not). We hope that larger portions of the energy and time currently invested in machine learning can be directed to making attribution possible, to assessing uncertainty correctly, and to achieving robust performance as to guide interventions. The agenda is clear and we have seen that promising steps have been taken. Another challenge is to educate the growing community of data scientists on what it takes to “learn” general truths from a series of individual observations. One hundred years of statistics (in which the “data modeling culture” has played a large role) have both exposed some of the challenges and identified a number of meaningful guarantees that mitigate the difficulties linked to inference. It is important that we effectively share this rich statistical thinking.

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Discussion of Paper by Brad Efron

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It is a privilege to have the chance of congratulating Professor Efron first on this wise paper, then on the richly merited International Prize, the award of which the paper commemorates, but above all on the whole body of his deeply impressive, wide-ranging contributions to our subject.

One issue which the paper indirectly raises is the role of different approaches to the conceptual and mathematical theory of our field. One of the appeals of the field is its totally international character yet, inevitably, broad and hazily defined national contrasts are visible.

Thus Professor Efron links “traditional” statistical theory to Neyman and Pearson. Yet the two set out to clarify earlier work of R.A. Fisher, first with Fisher’s encouragement, which only later turned to destructive hostility. The mathematical clarity of Neyman’s work is, of course, appealing but it may be argued that its overformalization continues to lead to misunderstanding, unproductive discussion and rigidity concerning, in particular, the role of significance tests.

The Nordic approaches have made an important and distinctive contribution to the field. See, in particular, the recent fine account of Rolf Sundberg, *Statistical modelling by exponential families*, stemming from the much earlier contribution of Per Martin-Lof. In the UK, Egon Pearson played a distinctive role in the development of our subject. He mostly had a preference for numerical illustration and was deeply involved in applications, stemming in part from an early visit he made to Bell Labs.

One of the great masterpieces of our field, largely unread nowadays, is M.S. Bartlett’s (1958) *Introduction to stochastic*

processes and their application and Bartlett’s influence on statistical development, at least in UK, was second only to Fisher’s. See, for example, his treatment in 1937 of asymptotic theory in which the dimension of the parameter space increases proportionally to sample size. One of the themes of his wide-ranging work was the use of specific stochastic processes, Markov chains, generalized birth-death processes and others, for the detailed interpretation of biological or physical science data; his final post was as Professor of Biomathematics. An aspect of both Fisher’s and Bartlett’s work, which I must admit I admire, is a total lack of concern with formal mathematical regularity conditions which, of course subject to due care, typically contribute little to understanding. Professor Efron in his theoretical papers manages with great skill and panache to combine careful mathematical discussion with judicious statistical emphasis. Special stochastic models probably tend nowadays to be treated much more often by computer simulation than by mathematical analysis of the differential equations involved with obvious gains and some loss, especially when the model is intended to lead to semi-qualitative understanding of an empirical phenomenon. The aethos is rather different from the use of families of regression-type models and even more so from that of machine learning, the latter achieving very specific purposes often with high effectiveness, as Professor Efron’s discussion so elegantly illustrates. But how useful are they as a guide to deeper interpretation or for broader context prediction?

I reemphasize my admiration for Professor Efron’s work and for this paper.



Comment: When Is It Data Science and When Is It Data Engineering?

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1. Introduction

The International Prize in Statistics (IPS) is awarded by the International Statistical Institute (ISI) every two years, to individuals whose work has made a seismic breakthrough in Statistics. The IPS rivals in stature to the Fields Medal in Mathematics, although it is more recent, and the ISI is to be highly commended for initiating the prize. Brad Efron's *bootstrap* stands out among all the amazing contributions made in Statistics in the last 50 years, and Brad himself is a singularity in the profession. He is known for his deep thinking about important science problems, elegant development of solutions with uncertainty quantified, and clear expositions of the statistical science. The IPS Laureate delivers a lecture at the World Statistics Congress (WSC). Brad's lecture at the 2019 WSC in Kuala Lumpur, Malaysia, was very forward-looking and concentrated on prediction (a more general concept than forecasting). Toward the end of my discussion, I move away from the specifics of Brad's very fine paper to a more philosophical place, where I suggest that prediction is looking more like Data Engineering than Data Science.

2. Prediction With an Extra Component of Variation

I really liked the way Brad trichotomized statistical analysis into prediction (inference on random quantities), estimation (inference on fixed but unknown parameters), and attribution (includes hypothesis testing, model choice, and classification). I would like to add to this, a fourth “-ion,” *decision* (in the presence of uncertainty), which I discuss in later sections.

Brad posed the prediction problem in terms of a “surface-plus-noise” paradigm, in which the surface is a latent process where the scientific truths reside. In his paper, he gives a decomposition that I go along with, until it stops short of what I need for spatial, temporal, and spatio-temporal prediction. In broad-brush strokes, his model for data y is

$$y = s(x, \beta) + \epsilon, \quad (1)$$

where $s(x, \beta)$ is a “surface” and ϵ is (independent) “noise.” I believe there is an extra component that should be included in (1), which is uncertainty in the representation of the scientific truths. That is, if the truths are represented by w (say), then

the degree of veracity of the surface $s(x, \beta)$ is captured by the equation,

$$w = s(x, \beta) + \xi, \quad (2)$$

which is the “process model” in a hierarchical statistical model (HM). Importantly, y are observations on w not on $s(x, \beta)$:

$$y = w + \epsilon, \quad (3)$$

which is the “data model” in an HM. Combining (2) and (3) results in:

$$y = s(x, \beta) + \xi + \epsilon, \quad (4)$$

which reduces to (1) if the process-model error, ξ , is identically zero. Interpreted in terms of linear regression, where $s(x, \beta) = x'\beta$, (4) tells us that the difference between the data and the surface is

$$y - x'\beta = \xi + \epsilon,$$

and hence if $x'\beta$ is a fixed effect the error budget is

$$\text{var}(y) = \text{var}(\xi) + \text{var}(\epsilon). \quad (5)$$

Misinterpretations of uncertainties are possible by not recognizing equation (5): The budget may not add up to the total variance if $\text{var}(\epsilon)$ is determined from independent calibration; or, if the noise-variance is assumed to be $\text{var}(y)$, over-smoothing will occur. In the model I use for spatial and spatio-temporal prediction, (4) is my starting point.

Usually, I want to predict w (where the scientific truths reside), for which the predictor,

$$\hat{w} \equiv E(w|y) = E(s(x, \beta)|y) + E(\xi|y),$$

is commonly used, along with its uncertainty, $\text{var}(w|y)$. In spatial, temporal, and spatio-temporal prediction, the process-model error ξ has statistical dependencies that contribute to both $E(w|y)$ and $\text{var}(w|y)$. Further, x could be random, a common example of which is the errors-in-variables problem. There are instances (e.g., when considering the output from numerical climate models) when it is the surface $s(x, \beta)$, not w , that one wants to predict. In this case, ξ is filtered out and one would use the predictor, $E(s(x, \beta)|y)$, with uncertainty, $\text{var}(s(x, \beta)|y)$.

The next section discusses how different queries about scientific truths in w usually lead to different optimal predictors.

3. Querying the Predictand

The *predictand* is the random quantity w (or some function of it) to be predicted. In spatio-temporal statistics, w is often a stochastic process that is observed incompletely and noisily, resulting in data y . A *predictor* of w is a statistic, namely a function of the data y that here we write as $\delta(y)$. We are doing this statistical analysis for a reason, presumably so that we can answer a question about how our world works.

In the presence of uncertainty, there is no universal optimal predictor of w that answers every question about it, which I now discuss briefly. A prediction can be looked upon as a *decision* in a decision-theoretic context, and hence predictions are made based on cost-benefit (utility, U) and risk (probabilities conditioned on the data y). There are many possible predictions δ , and we need some way to compare them. The utility tells us how well we do if the predictand is w , and we use δ to predict it; we write the utility as $U(w, \delta)$. We write the risk as $\Pr(w|y)$. Then weight the utility by the risk:

$$U(w, \delta) \Pr(w|y). \quad (6)$$

Recall that w is the random process we are trying to predict: In many of the problems I work on, w is a high-dimensional spatial or spatio-temporal random field for which I have incomplete and noisy data y . Then for each decision δ , $U(w, \delta)$ is an ensemble of utilities, where we can weight each one by its posterior probability (as we do in (6)). Integrating (6) yields the *posterior expected utility*,

$$\int U(w, \delta) \Pr(w|y) dw. \quad (7)$$

Then maximizing (7) with respect to δ over a given predictor space yields an optimal predictor δ^* that is a function of the data y .

Now suppose a different question is asked that involves making inference not on w but on a given function $g(w)$. A different utility function might be used, but suppose for the sake of argument that here it is the same. Then (6) is replaced with

$$U(g(w), \delta_g) \Pr(w|y), \quad (8)$$

where δ_g is a generic decision about $g(w)$. Integrating (8) yields,

$$\int U(g(w), \delta_g) \Pr(w|y) dw, \quad (9)$$

which when maximized yields δ_g^* that is a function of y . However, those who would like a simple life might base their prediction of $g(w)$ on

$$\tilde{\delta}_g(y) \equiv g(\delta^*(y)), \quad (10)$$

which is different from $\delta_g^*(y)$.

Now, if $\delta^*(y)$ is a great predictor of w , then would not $g(\delta^*(y))$ be a great predictor of $g(w)$? Yes, if there were complete certainty in w but, modulo death and taxes,

- our attempts to explain the world (science) are uncertain;
- our measurements of our uncertain world are uncertain.

For example, in the simple case where w is a random variable, the commonly used utility that has a squared-error penalty yields the optimal predictor, $\delta^*(y) = E(w|y)$, of w . Now consider prediction of $g(w) = \exp(w/a)$, where a is a known positive constant that has the same units as w . This problem occurs when, for example, soil or mineral properties are modeled on the log scale but inferences are needed back on the original scale. Then,

$$\delta_g^*(y) = E(\exp(w/a)|y) \neq \exp(E(w/a|y)) = g(\delta^*(y)),$$

because $g(\cdot) = \exp(\cdot/a)$ is nonlinear. In fact, because of Jensen's inequality,

$$\delta_g^*(y) \geq g(\delta^*(y));$$

and note that $\delta_g^*(y)$ is unbiased for $g(w)$. Hence, the “shape-shifting” predictor $g(\delta^*(y))$ is biased for $g(w)$, has inferior posterior expected utility, and should be avoided despite its one-size-fits-all appeal.

4. What Metrics Are We Using in the Twenty-First Century?

“What we measure affects what we do. If we have the wrong metrics, we will strive for the wrong things”: Joseph Stiglitz, Nobel Laureate in Economics (*Financial Times*, 13 September 2009).

Statistics has sometimes been described as a discipline that translates variability into uncertainty. We have all seen how our careful attempts to get the probability space just right, so that the uncertainty can be expressed with auditorial rigor, are ignored or misinterpreted by scientists and policy-makers. Really fast computers can ingest really big datasets and predict what look like really interesting signals. But are the signals real? Scientists (including statistical scientists) like to assess their predictions with the LG (Looks Good) “metric,” which works well when the signal is both strong and simple.

Statistical scientists can derive probabilities (denoted $\Pr(\cdot)$) for how different the predicted signal is from no signal. Statistics has “metrics” that include bias, variance, mean-squared (prediction) error, $\Pr(\text{Type I error})$, Power = $1 - \Pr(\text{Type II error})$, p -value, and so forth. In simulation experiments where the signal is known and from which the simulated data are generated, or in cross-validations, we can obtain empirical measures (or metrics), such as the false positive rate, false negative rate, false discovery rate, and false non-discovery rate.

For example, the false positive rate can be interpreted as an estimate of the conditional probability, $\Pr(\text{a signal is declared} | \text{no signal is present})$. Both the scientific method and the legal system are predicated on this type of probability, where there is a presumption of the *status quo* in Science or of innocence before the court. In statistical hypothesis testing, this is known as $\Pr(\text{Type I error})$, and in the legal system it corresponds to $\Pr(\text{conviction} | \text{innocent})$. Another metric is the false negative rate, where there is a presumption of a signal being present, that is, $\Pr(\text{no signal is declared} | \text{signal is present})$. In statistical

- our world is uncertain;

hypothesis testing, this is known as $\Pr(\text{Type II error})$, and in the legal system it corresponds to $\Pr(\text{acquittal} \mid \text{guilty})$.

To a software engineer, these metrics number among many that could be computed to evaluate an algorithm, but to a statistician the choice of metric depends on the question being asked. The statistician is trained to look for variability over sub-populations of the evaluated metric which, amongst other things, might indicate prejudicial behavior of an algorithm. However, our training typically comes up short when an answer is needed to the following type of question: Is a false positive rate or a false negative rate of 2% equally acceptable for a population of 250 people as it is for a population of 25 million? This is an example of where the decision-theoretic machinery in Section 3 could be used to establish cost-benefit in the presence of uncertainty and then used to quantify the choices available.

5. Robo-Debt: Australian Government Demands Its Citizens Prove Their Innocence

The Australian social-services agency, Centrelink, has data on the earnings of Australians receiving various types of welfare payments. Its records are not perfect, and it looks for data from other sources, such as the Australian Tax Office (ATO), to provide a cross-check. If there is a discrepancy, a please-explain notice is issued, at least that was the case up to the first half of 2016. In the middle of that year, a new scheme was introduced: It went from issuing about 20,000 notices annually to issuing 142,634 debt notices in the year following July 1, 2016, with a requirement that recipients prove there was no overpayment to them by Centrelink.

“Robo-debt,” as it was dubbed by the media, was devised under a conservative Australian government, where artificial intelligence (AI) was used to catch unentitled welfare recipients and human intelligence (HI) was all but suspended. Clause 39 of Magna Carta Libertatum (signed in 1215) recognizes individual rights over and above the arbitrary will of a government, and it formed the basis of that part of the Fourteenth Amendment of the US Constitution that guaranteed “due process of law” to its citizens. This right to due process has, over the last 800 years, included the presumption of innocence, and it is the cornerstone of most democracies. In Australia, from mid-2016 and beyond, that presumption was suspended in approximately 700,000 cases, using an algorithm that was flawed and a scheme that had a presumption of guilt at its core.

Robo-debt was introduced to save the government money by removing HI and relying on AI and “big data.” The robo-debt algorithm was based on matching the Centrelink database with an ATO database. An uneven wages profile, common in the gig economy, was treated differently by the two agencies, which prejudiced the casual and under-employed sector of the

economy. The debt notice that was sent included instructions that failure to pay would be pursued by private debt collectors or by garnishing the individual’s ATO tax returns. (I do not know of any credit notices being issued under the scheme.)

There was a right of appeal that involved a lengthy, cumbersome, and psychologically stressful process for a sector of the population that was already financially distressed. Moreover, even during the appeal, active collection of alleged debts proceeded. In media interviews, the government minister responsible quoted an approximately 1% error rate without clarifying exactly what that metric was. Subsequently, it was revealed that these “errors” were clerical data-entry errors and, in fact, the false positive rate of the scheme was on the order of 20%.

In the case of robo-debt, prediction failed because the algorithm was flawed and the metric was ill-defined. What the government did with that prediction was shocking and, in late 2019, the Australian Federal Court found the scheme to be illegal.

6. Science and Engineering

Broadly speaking, science tells us “why” and Engineering tells us “how.” We keep them separate in our academies and colleges. Is this a left-over from the Twentieth Century? In our daily life, do we no longer care why something works, as long as it works?

Up to now, Engineering has been an important vehicle for society’s uptake of Science’s discoveries. Machine Learning, which looks for patterns in very big datasets and classifies attributes according to those patterns, is often (incorrectly) equated with AI, which will allow us to do everything in our lives faster, smarter, and cheaper. . . until it will not because we have been incorrectly classified!

An AI algorithm is trained on a “typical” dataset, but it is almost certainly subject to unconscious bias that may or may not show up when applied to the general population. AI will tend to reinforce prejudices in the data in a manner similar to a Bayesian analysis where the posterior is used as a prior for the same or a similarly unrepresentative dataset. Are the error rates spread evenly across all demographics and, if not, is it because of unconscious bias?

In conclusion, Brad’s paper has embraced Data Science and forcefully argued for statistical thinking to be an essential component. The latter part of my discussion is purposely binary, so that as statisticians we continue to ask whether what we are doing is Data Science or Data Engineering.

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Discussion

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I congratulate Professor Efron on receiving the International Prize in Statistics. Although given primarily for his invention of the bootstrap, this and his many other contributions add up to a hugely influential body of work, and the award is richly deserved.

As a graduate student I had the privilege of hearing him speak in London, and as we left the seminar someone quietly said to me that this bootstrap thing could not possibly work. Many (too many!) years have passed since then, but what seemed visionary around 1980 has now come to pass: despite the initial doubts of some, resampling does indeed provide a very powerful approach to uncertainty assessment in a huge variety of settings, effectively replacing the analytical derivation of standard errors and suchlike with large-scale computations. There are caveats, of course, and during the 1980s and 1990s powerful theoreticians such as Peter Hall, sorely missed, shed light on how and when the bootstrap would work successfully and how to fix it when it failed. But the basic vision, that computer power could replace mathematical derivations, has come true. At the time some colleagues feared that computing power would replace statisticians, but this (to us) disastrous scenario has not materialized. The last time I looked, the top three “best jobs” in a standard US listing were “data scientist,” “statistician,” and “university professor” (very gratifying for those wearing all three hats), so I do not think we need fear the disappearance of “traditional” statistics any time soon.

One reason for this is that despite all the hype about scraping information from the web and other sources of big data, the vast majority of datasets remain small or moderate in size. Quite apart from issues of sampling bias and representativity, there is typically a quality/quantity tradeoff in data-gathering. One of many possible examples arises in studying the ages of those humans who live the longest, to see if there is an upper limit to the human lifespan. Individuals who live over 100 years form a tiny proportion of the general population, and even official documentation can be lacunary, so that a considerable effort is needed to verify their identities and thus their ages. One can have lots of data, mostly reliable, about the bulk of the population, but the information about the small group of most interest must be carefully verified. Moreover one runs up against the extrapolation problem mentioned in the paper—since the aspect of most interest lies outside the data, some sort of theory seems essential, be it mathematical, biological or both. Many problems are of this general sort, involving data that are of

doubtful value and with missing or incomplete scientific theory, and to which therefore so-called predictive methods may not apply.

I want to make two main comments about this interesting paper.

Prediction

My first comment is that I think we use the word “prediction” too glibly. Given sufficient data, linear regression or a deep learning, random forest or other machine learning algorithm could provide highly accurate predictions of when, and with what momentum, a stone dropped from the Leaning Tower of Pisa will hit the earth. It could not, however, help with a moon shot, since the force of lunar gravity is quite different, and learning in Pisa would not be relevant. Understanding of Newton’s laws, on the other hand, did enable a successful moon landing—which of course required enormous numbers of predictions that fortunately turned out to be correct. We should distinguish “algorithmic prediction,” based on empirical modeling of data, be it by linear regression, an ARIMA model or a deep learner, from “scientific prediction,” based on some knowledge, necessarily imperfect, of “laws of nature.” Only the latter enables the reliable transfer of knowledge from one setting to another—and prediction in a new context enables scientific theories to be tested, perhaps to destruction, or extended or modified. A further distinction to be made within “algorithmic” predictors is between those that provide some measure of the uncertainty of the prediction, which might properly be called “statistical,” and those, like most machine learning algorithms, that do not. So using logistic regression to obtain a single number would be an algorithmic prediction, whereas using it to give an estimated distribution would be a statistical prediction.

Although I would distinguish between scientific and empirical predictions, there are scientific settings in which algorithmic predictions might be appropriate. For example, numerical climate models, which are ultimately based on the Navier–Stokes equations, are becoming more and more detailed, but there will always be small-scale processes that they cannot properly include, either for computational reasons or because the processes are insufficiently well understood. Such processes might be better modeled empirically within a climate model driven by basic large-scale scientific considerations.

Uncertainty in Big Data

My second comment relates to independence assumptions in big data problems. In classical statistics we expect standard errors to shrink at rate $1/\sqrt{n}$, so when the sample size n is enormous everything becomes highly significant: for a huge sample, the logistic regression output shown by Professor Efron would have stars not only for every predictor but also for every possible interaction of the predictors, and we would find ourselves using the significance levels as a ranking tool rather than for attribution in Efron's sense. Yet it seems implausible that our usual over-simplified independence assumptions will apply to huge samples, owing to underlying factors that lead to subtle dependencies, individually too small to be detected, but important when cumulated. We are used to this in the familiar context of linear regression: the small dependencies between residuals can be neglected for many purposes, but when combined in the residual sum of squares, ignoring the loss of degrees of freedom can lead to serious underestimation of the error variance. As an aside, this is different from the correlations between predictors

that complicate attribution in wide-data problems. In genetic problems local correlations at the genome level mean that attribution to a single gene may be unreliable, whereas modeling correlations along the genome, in terms of hot spots, leads to more secure inferences. The correlations I refer to are between the rows of the data matrix and the response variables for different units. Sir David Cox, the first winner of the International Prize in Statistics, has discussed an approach to this based on notions of long-range dependence (Cox 2015), but surely much more remains to be done to ensure secure attribution in huge datasets, and some form of subsampling akin to the bootstrap might form the basis of a simple widely applicable approach.

To finish, I would like to congratulate Professor Efron again, and thank him not only for his stimulating paper, but also for all the tools he has given us.

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Discussion of “Prediction, Estimation, and Attribution” by Bradley Efron

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ABSTRACT

Professor Efron has presented us with a thought-provoking paper on the relationship between prediction, estimation, and attribution in the modern era of data science. While we appreciate many of his arguments, we see more of a continuum between the old and new methodology, and the opportunity for both to improve through their synergy.

Discussion

We would like to thank Professor Efron for a timely and thought-provoking paper. His discussion of the relationship between prediction, estimation, and attribution (a new term?) is an important one and deserves further thought.

Overall, however, we find ourselves only in partial agreement with his arguments. We summarize our views as follows:

1. Unlike Professor Efron, we do not see any fundamental tension between prediction and estimation/attribution. They have different goals, and both are needed. We agree that estimation and attribution may be harder than prediction. Science (e.g., biology) is much more complex than it was 200–300 years ago, and estimation and attribution are therefore more difficult. But in this complex world, prediction can still be successful and useful!

Here is an example from our own collaborative work (RT):

The project involves the diagnosis of skin cancer from 10,000 protein levels measured on skin biopsies. We collected training and test sets, and applied the lasso to try to distinguish cancer from normal tissue. The lasso chose about 200 proteins and predicts quite accurately (test set AUC about 95%). Now if we were to remove those 200 proteins from the dataset and refit, the new model would still predict well. This is because of correlation between predictors. Hence our model has not uncovered the ground truth of skin cancer: this is very difficult to do (it may take 10–20 years!). But the model can be used now to help doctors to diagnose skin cancer. Should we stop using it until we discover the ground truth? Of course not.

2. Professor Efron correctly states that the prediction problem can be “ephemeral.” But this does not distinguish it from estimation. Data and processes can drift. In estimation, the “law” that is true today may not hold exactly tomorrow. When modeling the spread of a virus like coronavirus, models for existing viruses are likely to be inadequate. Augmenting these models with empirical data seems like a crucial component

for accurate predictions. As Professor Efron notes, prediction tends to be applied more in ephemeral settings. This does not make it fundamentally more so.

3. We agree that interpretable prediction models may sometimes be helpful in getting closer to the ground truth. Simple interpretable models may sometimes be more robust and trustworthy. And when a simple model stops working, it is easier to diagnose and possibly correct the problem. Interpreting predictive models is an active area of research. Even when a prediction model appears to be working, it is important to know why.

For example, many groups are using deep learning to diagnose breast cancer from large sets of mammograms, with very promising results. In many cases the systems outperform expert radiologists (e.g., McKinney et al. 2020). However, sometimes the algorithms can be fooled by what had been called the “Clever Hans” effect (Lapuschkin et al. 2019). There can be a small label in the corner of the mammogram indicating the lab where the scan was taken. The network can seize on that label and exploit it to help classify the images. But this information is clearly not generalizable to the general population of mammograms. Hence, there is now a large body of work trying to pinpoint which pixels in a specific test image are responsible for the particular classification produced by a deep neural net.

4. In our view the modern prediction algorithms go hand-in-glove with the more traditional estimation models. Boosting for example can be viewed as estimation of a response surface in a surface plus noise scenario. If you fit a boosting model, and it explains 70% of the (test set) variance compared to the 30% explained by your parametric model, that suggests you may have left out some variables, some interactions, and possibly need some nonlinearities. On the other hand, if they do about the same, you have more confidence that your simpler explanation is on the right track. These same comments apply to other models, like logistic regression, Cox models, and so on, as well as to neural nets and Random Forests.

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Discussion of Professor Bradley Efron's Article on "Prediction, Estimation, and Attribution"

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1. Introduction

By noting the rapid growing trend of "pure prediction algorithms," Professor Efron compares and bridges the statistics of the 20th Century (estimation and attribution) to that of the current fast growing development of the 21st Century (prediction). The outstanding discussion offers many deep-rooted insights and comments. As did his forward thinking article on Fisher's influence on modern statistics (Efron 1998), which helped shape many recent developments on statistical inference (including our own work on confidence distribution (Singh, Xie, and Strawderman 2005; Xie and Singh 2013)), this equally inspiring article by Professor Efron will certainly galvanize many contemporary and powerful developments for modern statistics and for the foundations of data science.

In this note, we echo and also provide additional support to two important points made by Professor Efron: (1) prediction is "an easier task than either attribution or estimation"; (2) the IID assumption (e.g. random splitting of training and testing datasets) is crucial in the current developments on predictions, but we also need to do more for the case when the IID assumption is not met. Based on our own research, we provide additional evidence to support these discussions. We discover that prediction has a *homeostasis* property and works well under the IID setting even if the learning model used is completely wrong. We also highlight the importance of having a good modeling and inference practice: a good learning model with good estimation is important to improve prediction efficiency in the IID case and it becomes essential to maintain validity in the non-IID case. The message remains: we still need to make effort to build a good learning model and estimation algorithm in prediction, even if prediction is an easier task than estimation.

From the outset, we would like to point out that it is not a straw-man argument to consider non-IID testing data. On the contrary, such data are prevalent in data science. In addition to those examples provided by Professor Efron that showed "drift," we can easily imagine non-IID examples in many typical applications. For instance, a predictive algorithm is trained on a database of patient medical records and we would like to predict potential outcomes of a treatment for a new patient with more severe symptoms than what the average patient shows. The new patient with more severe symptoms is not a typical

IID draw from the general patient population. Similarly, in the finance sector, one is often interested in predicting the financial performance of a particular company. If a predictive model is trained on all institutes, then the testing data (of the specific company of interest) are unlikely IID draws from the same general population of the training data. The limitation of the IID assumption, in our opinion, has hampered our efforts to fully take advantage of fast-developing machine learning methodologies (e.g., deep neural network model, tree based methods, etc.) in many real-world applications.

Our discussions in this note are based on a so-called *conformal prediction* procedure, an attractive new prediction framework that is error (or model) distribution free (see, e.g., Vovk, Gammerman, and Shafer 2005; Shafer and Vovk 2008). We discover a homeostasis phenomenon that the expected bias caused by using a wrong model can largely be offset by the corresponding negatively shifted predictive errors under the IID setting. Thus, the predictive conclusion is always valid even if the model used to train the data is completely wrong. This robustness result clearly supports the claim that prediction is an easier task than modeling and estimation. However, the use of a wrong training (learning) model has at least two undesirable impacts on prediction: (a) a prediction based on a wrong model typically produces much wider predictive intervals than those based on a correct model; (b) although the IID case enjoys a nice homeostatic cancellation of bias (in fitted model) and shifts (in associated predictive errors) when using a wrong learning model, in the non-IID case this cancellation is often no longer effective, resulting in invalid predictions. The use of a correct learning model can help mitigate and sometimes solve the problem of invalid prediction for non-IID (e.g., drifted or individual-specific) testing data.

Section 2 reviews a conformal predictive procedure and shows that the prediction is valid under the IID setting, even if the learning model is completely wrong. Section 3 is a numerical study using a neural network model to demonstrate the impact of a wrong learning model and estimation on prediction in both the IID and non-IID cases. Section 4 is a concluding remark. A more detailed discussion, including an introduction of *predictive curve* (to represent predictive intervals of all levels) and an elaborated study of linear models, is in Xie and Zheng (2020).

2. Prediction, Testing Data, and Learning Models

As in Equation (6.4) of Professor Efron's article, we assume that a training (observed) dataset of size n , say, $\mathcal{D}_{\text{obs}} = \{(\mathbf{x}_i, y_i), i = 1, \dots, n\}$ is available, where $(\mathbf{x}_i, y_i), i = 1, \dots, n$, are IID random samples from an unknown population $\mathcal{F}$. For a given $\mathbf{x}_{\text{new}}$, we would like to predict what y_{new} would be. We first use the typical assumption that $(\mathbf{x}_{\text{new}}, y_{\text{new}})$ is also an IID draw from $\mathcal{F}$. Later we relax this requirement and only assume that $y_{\text{new}}|\mathbf{x}_{\text{new}}$ relates to $\mathbf{x}_{\text{new}}$ the same way as $y_i|\mathbf{x}_i$ relates to $\mathbf{x}_i$, but $\mathbf{x}_{\text{new}}$ follows a marginal distribution that is different from that of $\mathbf{x}_i$.

For notation convenience, we consider $(\mathbf{x}_{\text{new}}, y_{\text{new}})$ as the $(n+1)$ th observation and introduce the index $n+1$, with $\mathbf{x}_{n+1} = \mathbf{x}_{\text{new}}$ and y_{n+1} as a potential value of the unobserved y_{new} . Unless specified otherwise, the index " $n+1$ " and index "new" are exchangeable throughout the note.

2.1. Conformal Prediction Inference With Quantified Confidence Levels

The conformal prediction method has attracted increasing attention in learning communities in recent years (see, e.g., Vovk, Gammerman, and Shafer 2005; Shafer and Vovk 2008; Lei et al. 2018; Barber et al. 2019a, 2019b). The idea is straightforward. To make a prediction of the unknown y_{new} given $\mathbf{x}_{n+1} = \mathbf{x}_{\text{new}}$, we examine a potential value y_{n+1} , and see how "conformal" the pair $(\mathbf{x}_{n+1}, y_{n+1})$ is among the observed n pairs of IID data points $(\mathbf{x}_i, y_i), i = 1, \dots, n$. The higher the "conformality," the more likely y_{new} takes the potential value y_{n+1} . Frequently, a learning model, say $y_i \sim \mu(\mathbf{x}_i)$ for $i = 1, \dots, n, n+1$, is used to assist prediction. However, the learning model is not essential. As we will see later, even if $\mu(\cdot)$ is totally wrong or does not exist, a conformal prediction can still provide us a valid prediction, as long as the IID assumption holds.

To be specific, this note employs a conformal prediction procedure known as the *Jackknife-plus* method (see, e.g., Barber et al. 2019b). Consider a combined collection of both the training and testing data but with the unknown y_{new} replaced by a potential value y_{n+1} : $\mathcal{A} = \mathcal{D}_{\text{obs}} \cup \{(\mathbf{x}_{\text{new}}, y_{n+1})\} = \{(\mathbf{x}_i, y_i), i = 1, \dots, n, n+1\}$. We define *conformal residuals*

$$R_{ij} = y_i - \hat{y}_i^{-(ij)}, \quad \text{for } i \neq j \text{ and } i, j = 1, \dots, n, n+1,$$

where $\hat{y}_i^{-(ij)}$ is the prediction of y_i based on the leave-two-out dataset $\mathcal{A}^{-(ij)} = \mathcal{A} - \{(\mathbf{x}_i, y_i), (\mathbf{x}_j, y_j)\}$. If a working model $\mu(\cdot)$ is used, for instance, the model is first fit based on the leave-two-out dataset $\mathcal{A}^{-(ij)}$ and the point prediction is set to be $\hat{y}_i^{-(ij)} = \hat{\mu}(\mathbf{x}_i; \mathcal{A}^{-(ij)})$, where $\hat{\mu}(\cdot; \mathcal{A}^{-(ij)})$ is the fitted (trained) model using $\mathcal{A}^{-(ij)}$.

For each given y_{n+1} (a potential value of y_{new}), we define

$$Q_n(y_{n+1}) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{R_{n+1,i} \geq R_{i,n+1}\}}, \quad (1)$$

which relates to the degree of "conformity" of the residual values $R_{n+1,i} = y_{n+1} - \hat{y}_{n+1}^{-(i,n+1)}$ among the conformal residuals $R_{i,n+1} = y_i - \hat{y}_i^{-(i,n+1)}, i = 1, \dots, n$. (Here, $R_{i,n+1}$ are in fact the leave-one-out residuals of using the training dataset $\mathcal{D}_{\text{obs}}$.) If $Q_n(y_{n+1}) \approx \frac{1}{2}$, then $R_{n+1,i}$ is around the middle of

the training data residuals $R_{i,n+1}$ and thus "most conformal." When $Q_n(y_{n+1}) \approx 0$ or ≈ 1 , $R_{n+1,i}$ is at the extreme ends of the training data residuals $R_{i,n+1}$ and thus "least conformal." This intuition leads us to define a conformal predictive interval of y_{new} as

$$\begin{aligned} C_\alpha &= \left\{ y : Q_n(y) \geq \frac{\alpha}{2} \right\} \cap \left\{ y : 1 - Q_n(y) \geq \frac{\alpha}{2} \right\} \\ &= \left[q_{\frac{\alpha}{2}} \left(\{\hat{y}_{n+1}^{-(i,n+1)} + R_{i,n+1}\}_{i=1}^n \right), \right. \\ &\quad \left. q_{1-\frac{\alpha}{2}} \left(\{\hat{y}_{n+1}^{-(i,n+1)} + R_{i,n+1}\}_{i=1}^n \right) \right], \end{aligned} \quad (2)$$

where $q_\alpha(\{a_i\}_{i=1}^n)$ is the α th quantile of $a_1, \dots, a_n$. The interval (2) is a variant version of the *Jackknife-plus predictive interval* proposed by Barber et al. (2019b) in which $R_{i,n+1}$ is replaced by $|R_{i,n+1}| = |y_i - \hat{y}_i^{-(i,n+1)}|$ instead. The following proposition states that, under the IID assumption, C_α defined in (2) is a predictive set for y_{new} with guaranteed level- $(1 - 2\alpha)$.

Proposition 1. If $(\mathbf{x}_i, y_i), (\mathbf{x}_{\text{new}}, y_{\text{new}}) \stackrel{\text{iid}}{\sim} \mathcal{F}$, for $i = 1, \dots, n$, then $\mathbb{P}(y_{\text{new}} \in C_\alpha) \geq 1 - 2\alpha$.

A proof of the proposition can be found in Xie and Zheng (2020), which holds for a finite n . Barber et al. (2019b) pointed out empirically intervals like C_α have a typical coverage rate of $1 - \alpha$. In the rest of the note, we treat C_α as an approximate level- $(1 - \alpha)$ predictive interval.

Note that the function $Q_n(y)$ defined in (1) is in essence a *predictive distribution function* of y_{new} (see, e.g., Shen, Liu, and Xie 2018; Vovk et al. 2019). The corresponding *predictive curve* of y_{new} is

$$\text{PV}_n(y) = 2 \min\{Q_n(y), 1 - Q_n(y)\}.$$

Clearly, $C_\alpha = \{y : \text{PV}_n(y) \geq \alpha\}$. A plot of predictive curve function $\text{PV}_n(y)$ provides a full picture of conformal predictive intervals of all levels. Analogous to that of confidence distribution and Birnbaum's confidence curve, predictive function $Q_n(y)$ has a confidence interpretation as the p -value function of the one-sided test $H_0 : y_{\text{new}} = y$ versus $H_1 : y_{\text{new}} \leq y$, and the predictive curve $\text{PV}(y)$ has the same implementation for the corresponding two sided test (see, e.g., Xie and Zheng 2020, sec. 2.2). A formal definition of conformal predictive function is in Vovk et al. (2019).

A striking result is that Proposition 1 holds, even if the learning model $\mu(\cdot)$ used to obtain the prediction is completely wrong, as long as the IID assumption holds. This robust property against model misspecification is highly touted in the machine learning community. It gives support to the sentiment of using "black box" algorithms where the role of model fitting is reduced to an afterthought, although we will also provide arguments to counter this sentiment.

2.2. IID Versus Non-IID: Efficiency and Validity Under a Wrong Model

Although the validity of prediction is robust against wrong learning models in the IID case, there is no free lunch. The predictive intervals obtained under a wrong model are typically

wider. For instance, suppose that the true model is $y = \mu_0(\mathbf{x}) + \epsilon$, but a wrong model $y = \mu_1(\mathbf{x}) + e$ is used. Since $y = \mu_0(\mathbf{x}) + \epsilon = \mu_1(\mathbf{x}) + \{\mu_0(\mathbf{x}) - \mu_1(\mathbf{x})\} + \epsilon$, we have $e = \{\mu_0(\mathbf{x}) - \mu_1(\mathbf{x})\} + \epsilon$. So, when ϵ is independent of $\mathbf{x}$, $\text{var}(e) = \text{var}(\{\mu_0(\mathbf{x}) - \mu_1(\mathbf{x})\}) + \text{var}(\epsilon) \geq \text{var}(\epsilon)$ and the equality holds only when $\mu_1(\mathbf{x}) = \mu_0(\mathbf{x})$. Thus, the error term e under a wrong model has a larger variance than the error term ϵ under the true model. The larger the variance $\text{var}(\{\mu_0(\mathbf{x}) - \mu_1(\mathbf{x})\})$ is (i.e., the more discrepant $\mu_1(\mathbf{x})$ and $\mu_0(\mathbf{x})$ are), the larger the variance of the error term e is. A larger error translates to less accurate estimation and prediction. See also Proposition 2 of Xie and Zheng (2020) for a formal statement regarding the predictive interval lengths in linear models.

We have an explanation why a conformal predictive algorithm can still provide valid prediction even under a totally wrong learning model in the IID case. Specifically, when we use a wrong model $\mu_1(\mathbf{x})$, the corresponding point predictor will be biased by the magnitude of $\mu_1(\mathbf{x}_{\text{new}}) - \mu_0(\mathbf{x}_{\text{new}})$, but at the same time the error term e absorbs the *bias* and produces residuals with a *shift* by the magnitude of $\mu_0(\mathbf{x}_i) - \mu_1(\mathbf{x}_i) = -\{\mu_1(\mathbf{x}_i) - \mu_0(\mathbf{x}_i)\}$. In the conformal predictive interval (2), the quantiles of residuals are added back to the point prediction to form the interval bounds. If the IID assumption holds, the bias is offset by the shift. See also Xie and Zheng (2020) in which an explicit mathematical expression of this cancellation in linear models is derived. Along with greater residual variance, the offsetting ensures the validity of the conformal prediction in the IID case. We call this tendency of self-balance to maintain validity a *homeostasis phenomenon*.

The IID assumption is a crucial condition to ensure the validity of a prediction under a wrong model. If the IID assumption does not hold for the testing data, the prediction based on a wrong learning model (or a correct model but a wrong parameter estimation) is often invalid with large errors, as we see in our case studies. We think this IID assumption also explains why deep neural network and other machine learning methods work so well in academic research settings (where random split of data into training and testing sets is a common practice) but fail to produce “killer applications” to make predictions for a given patient or company whose $\mathbf{x}_{\text{new}}$ are often not close to the center of the training data. The good news is that, if we use a correct model for training and can get good model estimates, it is still possible to get a valid prediction for a specific $\mathbf{x}_{\text{new}}$. Modeling and estimation

remain relevant and often crucial for prediction in both IID and non-IID cases.

3. Case Study: Prediction Under Neural Network Models

We use a neural network model and a simulation study to provide an empirical support for our discussion. In the current neural network development, model fitting algorithms do not pay much attention to correctly estimate the model parameters. We find that the estimation of model parameters also plays an important role in prediction, in addition to a correct model specification.

Example 1. Suppose our training data $(y_i, \mathbf{x}_i)$, $i = 1, \dots, n$, are IID samples from the model

$$\begin{aligned} y_i &= \mu_0(\mathbf{x}_i) + \epsilon_i \\ &= \max\{0, \max\{0, z_{i1} + z_{i2}\} - \max\{0, w_i\}\} + \epsilon_i, \\ \epsilon_i &\stackrel{\text{iid}}{\sim} N(0, \sigma^2), \end{aligned} \quad (3)$$

where $\mathbf{x}_i = (z_{i1}, z_{i2}, w_i)^T \stackrel{\text{iid}}{\sim} N(\mu_x, \Sigma_x)$ and ϵ_i and $\mathbf{x}_i$ are independent. Here, $\mu_x = (0, 0, 0)^T$, the (k, k') -element of Σ_x is $0.5^{|k-k'|}/2$, for $k, k' \in \{1, 2, 3\}$, $\sigma^2 = 1$ and $n = 300$. Model (3) is in fact a neural network model (with a diagram presented in Figure 1(a)) and we can re-express $\mu_0(\mathbf{x}_i)$ as

$$\mu_0(\mathbf{x}_i) = f(A_2 f(A_1 \mathbf{x}_i)). \quad (4)$$

Here, $f(x) = \max(x, 0)$ is the ReLU activate function, and $A_1 = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \end{pmatrix}$ and $A_2 = (a_1^{(2)}, a_2^{(2)})$ are the model parameters. Corresponding to (3), the true model parameter values are $a_{11}^{(1)} = a_{12}^{(1)} = a_{23}^{(1)} = 1$, $a_{13}^{(1)} = a_{21}^{(1)} = a_{22}^{(1)} = 0$ and $(a_1^{(2)}, a_2^{(2)}) = (1, -1)$. In our analysis, we assume that we know the model form (4) but do not know the values of model parameters A_1 and A_2 .

For the testing data, we consider two scenarios: (i) [IID case] $\mathbf{x}_{\text{new}} \stackrel{\text{iid}}{\sim} N(\mu_x, \Sigma_x)$ and, given $\mathbf{x}_{\text{new}}$, $(\mathbf{x}_{\text{new}}, y_{\text{new}})$ follows (3); (ii) [Non-IID case] the marginal distribution $\mathbf{x}_{\text{new}} \stackrel{\text{iid}}{\sim} (T_1, T_2, T_3)$ and, given $\mathbf{x}_{\text{new}}$, $(\mathbf{x}_{\text{new}}, y_{\text{new}})$ follows (3). Here, T_1, T_2, T_3 are iid random variables from the t -distribution with degrees of freedom 3 and non-centrality parameter 1.

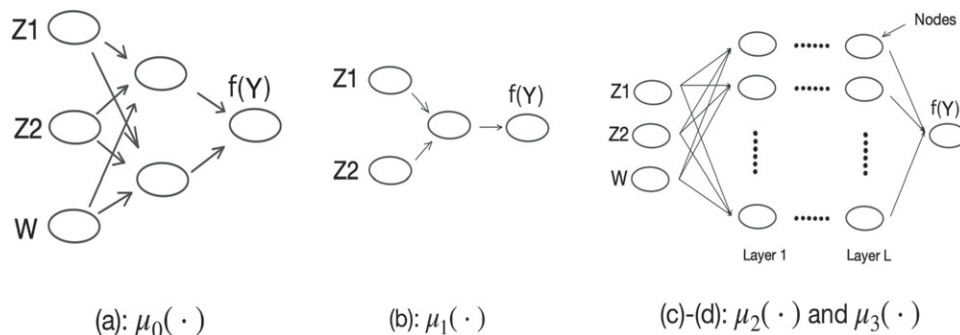


Figure 1. Diagrams of four neural network models: (a) true $\mu_0(\cdot)$; (b) partial $\mu_1(\cdot)$; and (c, d) over-parameterized $\mu_2(\cdot)$ and $\mu_3(\cdot)$ of (20 nodes in each layer) $\times L$ layers, with $L = 20$ and 100, respectively.

Table 1. Mean square error of each parameter in μ_0 (training data $n = 300$; repetition = 10).

MSE	a_{11}	a_{12}	a_{13}	a_{21}	a_{22}	a_{23}	b_1	b_2
Opt-MSE	0.07	0.059	0.31	0.154	0.109	0.124	0.06	0.101
Neuralnet	4.87	5.9	1.53	1.14	2.23	2.08	4.87	0.84

In addition to (a) the true model $\mu_0(\cdot)$, four wrong learning models are considered:

- (b) $\mu_1(\mathbf{x}_i) = f(B\mathbf{z}_i)$ (partially correct neural network model, missing w_i);
- (c) $\mu_2(\mathbf{x}_i) = f(C_{20}f(C_{19} \cdots f(C_1\mathbf{x})))$ (deep neural network model with 20 layers);
- (d) $\mu_3(\mathbf{x}_i) = f(D_{100}f(D_{99} \cdots f(D_1\mathbf{x})))$ (deep neural network model with 100 layers);
- (e) $\mu_4(\mathbf{x}_i) = \eta_0$ (without any covariates),

where $\mathbf{z}_i = (z_{i1}, z_{i2})^T$, $B = (b_1, b_2)$, $C_1, D_1 \in \mathbb{R}^{20 \times 3}$, $C_{20}, D_{100} \in \mathbb{R}^{1 \times 20}$, and $C_i, D_j \in \mathbb{R}^{20 \times 20}$, $2 \leq i \leq 19$, $2 \leq j \leq 99$. In our analysis, the neural network models $\mu_0(\cdot) - \mu_3(\cdot)$ are fit using the NEURALNET package (cran.r-project.org/web/packages/neuralnet/).

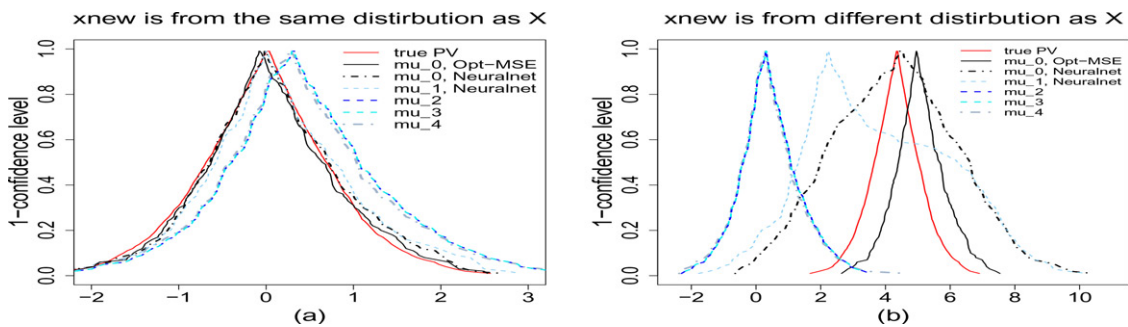
The Neuralnet package is an off-the-shelf machine learning algorithm. Its emphasis is on learning and not on model parameter estimation. Even under the true model $\mu_0(\cdot)$, the estimates of model parameters from Neuralnet are not very accurate (see Table 1). In the table, “Opt-MSE” refers to a code that we wrote by directly minimizing $\text{MSE} = \sum_{j=1}^n (y_j - \mu_0(\mathbf{x}_j))^2$, which can be implemented when the neural network is small. The calculation in the table is based on 20 repeated runs, each with a training dataset of size $n = 300$ from model (3).

Reported in Table 2 are the coverage rate and average interval length of predictive intervals computed under $10 = 5 \times 2$ settings with five different learning models $\mu_k(\cdot)$, $k = 0, 1, \dots, 4$, and in two scenarios. The analysis is repeated for 10 times with 10 simulated training datasets from model (3). We use 10 repetitions and not a greater number, because it takes a long time to fit a neural network model. However, for each of the 10 training datasets, 20 pairs of $(y_{\text{new}}, \mathbf{x}_{\text{new}})$ are used. So, for the reported values, each is computed using $10 \times 20 = 200$ pairs of $(y_{\text{new}}, \mathbf{x}_{\text{new}})$. For the true neural network model $\mu_0(\cdot)$, Opt-MSE is also used to fit the model. As we can see in Table 2, under the IID scenario, all predictive intervals are valid with a correct coverage. The best one with the shortest interval length is the one that uses the correct model and Opt-MSE estimation method. In the non-IID case, only the shallow neural network models provide valid predictions, and among them, Opt-MSE can give us confidence intervals with half the width. Indeed, when a wrong learning model is used, the IID assumption is essential for the prediction validity and the use of a wrong model often results in wider intervals. Furthermore, the estimation of model parameters seems to also have a big impact on prediction.

To get a full picture of the predictive intervals at all levels, we plot in Figure 2 the predictive curves of y_{new} . The plots are based on the first training dataset and making prediction for (a) the IID case with the realization $\mathbf{x}_{\text{new}} = (-0.909, -1.149, -0.771)$, and (b) the non-IID case with the realization $\mathbf{x}_{\text{new}} = (3.653, 1.748, 1.063)$. The realized value of $\mu_0(\mathbf{x}_{\text{new}})$ is 0 and 4.338 in (a) and (b), respectively. From Figure 2, we see that the use of a wrong model $\mu_1(\cdot) - \mu_4(\cdot)$ results in wider predictive curve (and predictive intervals at all levels $1 - \alpha \in (0, 1)$) in both IID and non-IID cases. Although the shallow neural network models $\mu_0(\cdot)$ and $\mu_1(\cdot)$ can provide good coverage rates, the predictive curves in the non-IID case are much wider than other approaches. This peculiar phenomenon occurs even when we assume to know

Table 2. Performance of 95% predictive intervals under five different learning models and in two scenarios: coverage rates (before brackets) and average interval lengths (inside brackets) (training data size = 300; testing data size = 20; repetition = 10).

	True model		Wrong model			
	$\mu_0(\cdot)$		$\mu_1(\cdot)$	$\mu_2(\cdot)$	$\mu_3(\cdot)$	$\mu_4(\cdot)$
	Opt-MSE	Neuralnet	Nueralnet	Nueralnet	Nueralnet	Nueralnet
IID scenario	0.995 (4.462)	0.99 (4.608)	0.99 (4.809)	0.99 (5.212)	0.99 (5.201)	0.985 (5.26)
Non-IID scenario	0.955 (4.52)	0.985 (9.327)	0.98 (9.77)	0.71 (5.899)	0.695 (5.201)	0.685 (5.277)

**Figure 2.** Plots of predictive curves for (a) $\mathbf{x}_{\text{new}} \stackrel{\text{iid}}{\sim} \mathbf{x}_i$ and (b) $\mathbf{x}_{\text{new}} \not\sim \mathbf{x}_j$. In each plot, the red solid curve is the target (oracle) predictive curve $PV_n(y) = 2 \max\{\Phi(y - \mu_{\text{new}}), 1 - \Phi(y - \mu_{\text{new}})\}$, obtained assuming that the distribution of $y_{\text{new}} \sim N(\mu_{\text{new}}, 1)$ is completely known. The two predictive curves obtained using $\mu_0(\cdot)$ are in black (solid line for Opt-MSE; dashed line for Neuralnet). The other predictive curves (all in a dashed or broken line and in various colors) are obtained using the other four wrong working models.

the true model structure $\mu_0(\cdot)$, indicating the importance of estimating model parameters accurately. Furthermore, in the non-IID case, there are large shifts when we use deep neural network models $\mu_2(\cdot)$ and $\mu_3(\cdot)$, leading to invalid predictions. The best prediction result is from the one obtained by using the correct learning model $\mu_0(\cdot)$ with the more accurate parameter estimation method Opt-MSE. The message is the same as what we have learned from Table 2, which also exactly mirrors what is found in the case study of linear models in Xie and Zheng (2020).

4. Conclusion

Professor Efron pointed out that “the 21st Century has seen the rise of a new breed of what can be called ‘pure prediction algorithms.’” We are fully in agreement with Professor Efron’s discussion that the prediction algorithms “can be stunningly successful,” and that “the emperor has nice clothes but they’re not suitable for every occasion.” Along the same line and under the setting of conformal prediction, we have demonstrated and explained how and why a prediction method can be successful under the IID assumption, even if the learning model is completely wrong. More importantly, we have also demonstrated that it is still meaningful, and often crucial, to build our prediction algorithms based on a good practice of modeling, estimation and inference. We fully anticipate and believe that “the most powerful ideas of Twentieth Century statistics”—modeling, estimation, and inference, will play a pivotal role in building the mathematical foundation of modern data science and in fully realizing its potential for real-world applications.

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The Data Science Process: One Culture

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We would like to congratulate Professor Brad Efron for the well-deserved honor of winning the 2019 International Statistics Prize, and for his thought-provoking paper discussing the diverging paths of so-called traditional statistical regression and pure prediction approaches. Professor Efron has made vast contributions to the field of statistics spanning several decades, and we enjoyed reading his take on the current state of the fields of statistics and machine learning.

In this discussion, we provide a portrayal of our own perspectives on the intersection of statistics and machine learning methods, and compare them with Professor Efron's. While we came from very similar beginnings as Professor Efron, rather than arriving at his view of a “discrepancy” or a “radical difference” between “traditional statistical regression methods” and “pure prediction” methods, we arrived at a surprisingly contrasting view that unifies these two approaches. At the very end of his paper, Professor Efron seems “hopeful” about the “reunification” of these two paradigms. However, in his view, efforts toward this reunification are “just underway,” with the primary impediments being a lack of theory. In our contrasting view, we are much further along the path of reunification, with the theoretical underpinnings being less critical than—but still important, and following on from—empirical evidence in today's reality-rooted era.

While our views are quite different, we certainly agree with many of the points that Professor Efron makes in his paper. For instance, Professor Efron appears weary of relying solely on *random* splits when implementing the train/test paradigm that is pervasive in the world of pure prediction algorithms. Specifically, he points out that there are instances where instead of providing an honest estimation of the error, a random split will be far too optimistic, as in the case of time-dependent data. He argues that a more realistic split in such scenarios is not random, but rather uses an early/late split where the early data are used to train and the later data are used to test. Indeed, a random split in many cases ensures that the training and test sets are identically distributed, but fails to ensure that they are independent (i.e., a random split does not resemble two independent samples taken from the same population). Such a random split certainly does not resemble the relationship between the data used to train an algorithm and the future data that it will be used on (future data are, at best, an independent sample from the same population, but are often collected from a different source

or under different circumstances). While Professor Efron uses this example to focus on how prediction is easier than extrapolation or interpolation (based on the observation that a random split yields better test-set performance than the early/late split in time-dependent data), we view this as an example of the importance of critical thinking. In our view, the train/test paradigm should not focus on randomness, it should instead focus on representing the way the predictive algorithm will be used. The training set should be representative of the current data being used to train the algorithm, and the test set should be representative of the future data to which the algorithm will be applied. So while we agree with Professor Efron's point that a random split is not always appropriate, we argue that it is not the *randomness* of the split that is critical to the pure prediction paradigm, but the *representativeness* of the split (relative to the prediction problem).

We were particularly delighted to read Professor Efron's reflections on Leo Breiman's 2001 two cultures paper, including his own discussion and perspectives at the time, along with how they have been updated in the two decades since. Interestingly, Leo's paper, and author Yu's own interactions with him, were the starting point for her own foray into machine learning 20 years ago. The lessons learned and perspectives gained from this journey have become a focal part of Yu's own research and philosophy that she has both passed on to—and developed together with—her students (including coauthor Barter). Today looking back on Breiman's work, the main difference between our view and Professor Efron's is that his view focuses on the differences between machine learning (or “pure prediction”) methods and “traditional regression” methods, whereas our own view focuses on the connections that bridge these two cultures, and the expansions building on both.

Professor Efron highlights his perspective on the differences between “pure prediction” and “traditional regression methods” via six criteria presented in Table 5. For instance, he views traditional regression models as focusing on scientific truth with pure prediction methods focusing instead on prediction accuracy; he views traditional regression models as focusing on low dimension problems while pure prediction methods focus on high dimensional problems; and he views traditional regression problems as utilizing optimal inference theory such as maximum likelihood and Neyman–Pearson while pure prediction methods focus instead on training/test paradigms.

However, in our experience, pure prediction methods are routinely used in practice *together* with classical statistical methods such as PCA and logistic regression. Right or wrong, logistic regression is often even categorized as a machine learning (pure prediction) method, and many modern machine learning books include traditional statistical ideas such as maximum likelihood, linear regression, and logistic regression. We have experienced the use of both traditional regression and pure prediction approaches in low- and high-dimensional problems. The idea that traditional regression methods focus on scientific truth while pure prediction approaches focus on prediction accuracy certainly has some merit, but in our experience the “truth” that traditional regression methods supposedly represent is rarely justified or validated. The mathematical formulae on which traditional regression models are built are rarely based on verifiable domain knowledge. Instead, the primary method for which people find evidence that these mathematical regression formulations represent the truth is often measured by their *predictive accuracy* (which, in Professor Efron’s paper, is the primary concern of pure prediction methods, rather than traditional regression models). At the very least, if traditional regression methods do not generate accurate predictions, then it is unlikely that it is adequately capturing any kind of underlying “truth.” Moreover, since a reasonable goal is to ensure that conclusions drawn (and predictions generated) are generalizable to new data, the training/test paradigm is just as relevant for traditional regression approaches as it is for the pure prediction approaches.

Professor Efron separates regression approaches into three categories: prediction (“the prediction of new cases”), estimation (“an instrument for peering through the noisy data and discerning a smooth underlying truth”), and attribution (“the assignment of significance to individual predictors”).

Professor Efron argues that pure prediction algorithms tend to “focus on prediction, to the neglect of estimation and attribution.” However, while traditional regression methods generate attribution via *p*-values associated with model coefficients, many pure prediction methods, such as random forest, have their own methods of attribution in the form of variable importance (gini or permutation) scores. Professor Efron discusses this idea but dismisses these modern attribution attempts as being suboptimal, dampened by the pure prediction method’s focus on prediction using many weak learners that do not prioritize individual strong predictors. Meanwhile, in our own research group we have seen great success using these importance scores to understand attribution. We developed the iterative random forest (iRF) (Basu et al. 2018) approach for identifying higher-order interactions based on decision paths in (pure prediction) random forest models, and have found such approaches to be extremely useful in practical applications including verifying existing and identifying novel gene interactions in drosophila embryonic development. Our experience indicates that prediction and attribution can go hand-in-hand, rather than being at odds with one another. In addition, Professor Efron’s focus on estimation implies that there is an underlying “truth” that can be uncovered by a model. As we discussed above, our view has become that no such truth is ever really attainable, whether or not you are working within the realm of traditional regression models or pure prediction

models. Our own perspective has transitioned into a framework of accurate *approximation* for a particular domain, and relative to a particular performance metric, rather than *truth-seeking* as was the goal of traditional statistical inference.

Finally, Professor Efron calls for “substantial further development” of machine learning approaches before they are ready for “routine scientific applicability.” However, such approaches are already being used successfully in routine scientific applicability. Of course we also agree that further development is always a good thing. In our research group, we have been working toward this goal, building on our own experience gained from many interdisciplinary projects applying pure prediction methods in conjunction with traditional statistical ideas in neuroscience, genomics and precision medicine. As a result of our work, we have developed a framework that we call *veridical* (“truthful”) *data science*, which encompasses a unified view of pure prediction and traditional regression approaches under the three principles of predictability, computability, and stability (PCS). The idea of using prediction as a reality check forms the basis of our predictability principle, and ideas from the traditional regression approaches are modernized in the stability principle, which itself bears a strong relationship to and expands on Professor Efron’s groundbreaking work on the bootstrap. Stability is a critical component of all stages of a data science project, ranging from the literal words used to formulate the problem (every word in the problem formulation should mean the same thing—have stable meaning—to every individual on a multidisciplinary project team) to data cleaning, and data and model perturbations in the modeling stage. Data-driven approaches are brought into the modern big data era via the computability principle. Our “one culture” veridical PCS data science framework highlights the practice of veridical (“truthful”) data science for reliable, reproducible and transparent data-driven decision making and knowledge generation from data for a particular domain problem (Yu and Kumbier 2020). In addition, our paper also introduces and advocates for PCS documentation which records the analytical judgement calls made by humans throughout the data science life cycle, and records relevant domain knowledge and code. Our paper provides a nice summary of our views as this one does for Professor Efron’s. The two papers would be an interesting side-by-side read.

Professor Efron’s piece gave us much food for thought, allowing us the opportunity to reflect on changing perspectives over time. We appreciate the invitation to provide a discussion for this piece, and we remain ever in debt to the truly innovative contributions that Professor Efron has made to the field of statistics that have now also made their way into the field of machine learning, and in our view, contributed to the unification (rather than the division) of these two fields to arrive at “one culture” in today’s field of data science.

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Rejoinder

Bradley Efron

A hot new statistical technology like the pure prediction algorithms divides the discipline into insiders and outsiders. The insiders, caught up in the excitement of new lands to conquer, are usually in no mood to think critically about methodological weaknesses. Outsiders, everyone else, may lack enough information and insight to mount an effective critique, especially if the new methods are intricate, sprawling, and somewhat mysterious. Eventually though, there is enough mixing between the two groups to allow the critical process to begin. Statistics has a strong tradition of internal criticism—witness the Bayesian/frequentist debates—with discussion papers like this one serving as useful outlets.

The discussants here span the insider/outsider spectrum, including some of the most prominent developers of modern prediction methods. Predictably, the insiders are the happiest with the fit between traditional and algorithmic methods: they “go hand in glove” say Professors Friedman, Hastie, and Tibshirani, (henceforth FHT), speaking with authority as the authors of seminal texts on statistical learning, as well as key technologies such as the lasso. “Data Science Process: One Culture” is the title of Professor Yu and Dr Barter’s commentary, a reverse echo of Leo Breiman’s famous “two cultures” paper, which, contrariwise, claimed a stark dichotomy between traditional methods and the “prediction culture.” Again, Yu/Barter speak as experienced and accomplished inside developers of modern prediction methodology.

The outsiders are less happy with the current state of connection between traditional and algorithmic regression ideas. (The term “outsider” here is only relative to algorithmic prediction; all of the discussants have been influential developers of modern data-analytic techniques such as false discovery rates, genomic analysis, and the bootstrap.) This paper was written from an outsider’s point of view. Rereading it now, a year later, made me aware of some outsider limitations: there is in fact more connective tissue between the two cultures than I credited, both FHT and Yu/Barker giving convincing examples.

That being said, an extra year of experience has not changed my belief in a disconnect between traditional statistical regression methods and the pure prediction algorithms. Section 8 of the paper, featuring Table 5, makes the case directly in terms of six criteria. Five of the six emerged more or less unscathed from the discussion. Criteria 2, long-time scientific truth versus possibly short-term prediction accuracy, was doubted by FHT and received vigorous push-back from Yu/Barter:

... but in our experience the “truth” that traditional regression methods supposedly represent is rarely justified or validated...

This is a hard-line Breimanian point of view. That “rarely” is over the top. The truism that begins “all models are wrong” ends with “but some are useful.” Traditional models tend to err on the side of over-simplicity (not enough interactions, etc.) but still manage to capture at least some aspect of the underlying mechanism. “Eternal truth” is a little too much to ask for, but in the Neonate example we did wind up believing that respiratory strength had something lasting to do with the babies’ survival.

Beyond the assessment of a model’s correctness lies a deeper question: do we believe that there *is* a model? The history of science is a story of successful models—Newton, Maxwell, Einstein, circulation of the blood, the periodic table, seafloor spreading, DNA, and the double helix. Prediction by itself is an end state. Often a very useful end state, but we seem to need some form of theoretical underpinning to make further scientific progress. This is the place where the paper parts ways with the hard-line predictionist point of view.

A side effect of the anti-modeling ethos is the ad hoc nature of the pure prediction algorithms; ingeniously ad hoc it is fair to say, but still devoid of theoretical justification, as shown for instance by the disparity between random forests and deep learning. The fathers of statistical theory—Pearson, Fisher, Neyman, Hotelling, Wald—forgot to provide us with a comprehensive theory of optimal prediction. We will have to count on the current generation of young statisticians to fill the gap and put prediction on a principled foundation.

Professor Cressie adds a fourth category, “decision,” to my prediction-estimation-attribution triad, and also adds a term to the surface-plus-noise paradigm, allowing the underlying “truth” to vary over time. He makes an interesting comparison between data science and data engineering, not unlike the paper’s distinction between long-term and ephemeral inferences, science being more “why” and engineering more “how.” The Robo-debt story is a chilling example of what can go wrong when using the wrong metric of prediction accuracy (too much “how.”) The question of prediction metrics, missing from the paper, is also raised by Professors Candes and Sabatti.

Candes and Sabatti’s wide-ranging essay brings up a variety of interesting topics and questions:

- I used “attribution” to avoid “significance,” an iffy terminology in general, and definitely out of place here.

- Causal inference, along with reproducibility another hot topic, lives on a far shore from the prediction algorithms, being a model-heavy topic. But, as Candes/Sabatti say, there are ongoing attempts to use the algorithms for causality purposes. Maybe this would have been an apt example for both of Section 10's "hopeful trends."
- In de Finetti's profound Bayesian universe, parameters (let alone parametric models) do not exist, only observations and how well we can predict them. Candes/Sabatti smartly connect this with Yu and Kumbier's extensively developed theory of "veridical" data science.
- The simplest probability model for prediction problems is the independence of the (x, y) pairs from one another in (2.1). How far can independence by itself take us? The non-parametric bootstrap is one example, conformal inference another, and also, almost, the ingenious knock-off methods Candes/Sabatti describe. It is worth remembering that the independence model *is* a model and can go wrong, as the discussion of concept drift in Section 6 shows.

This last point connects with Professor Xie and Zheng's discussion, a short paper in itself. Their Proposition 1 shows that conformal inference methods produce well-calibrated prediction intervals in surface plus noise models under an iid assumption for the (x, y) pairs, even if the assumed surface model is wrong. In the cholestyramine example of Figure 1, for instance, we might assume a cubic regression even if it should actually be eighth degree. The penalty for using the wrong model in the conformal calculations is increased width of the calculated prediction intervals. Proposition 1 is a more sophisticated version of the fact that the nonparametric bootstrap gives correct standard errors under iid sampling, even for "incorrect" estimators, that is, the variability of Figure 1's cubic regression curve whether or not the true model is cubic.

Professor Davison brings up the question of sample size n . In his practice, and much of mine too, n still tends to be small, whereas, as Candes/Sabatti point out, in the prediction world n seems actually to be going to infinity, or at least approaching the $n = \text{all ideal}$, where traditional stalwarts such as $\sqrt{n}$ accuracy, no longer apply. My own experience has followed the "too big, too small" principle: the investigator brings in a messy mountain of data, but by the time I figure out the right question to ask, the relevant part of the mountain shrinks to a molehill. I cannot apologize very much for the small size of the paper's examples. Huge sample sizes can conceal a method's weaknesses, which become worrisome when one gets to the molehill stage.

ARIMA models were a hot topic when introduced in the early 1970s, and, like the prediction algorithms, they carried with them a somewhat mysterious allure. It took a decade of critical thinking to reduce the mystery and put ARIMA into its place as a well-understood standard statistical tool. My hope is that the same process is now under way for prediction algorithms.

Nobody has done more than Professors Friedman, Hastie, and Tibshirani to connect the statistics and prediction algorithm communities, through their work and, especially, through their books. I will try to follow their comment numbering in what follows:

- (1) I certainly agree that prediction is a useful part of statistical practice. The paper's critique was aimed at the pure prediction algorithms, random forests etc., and not at prediction in general. Lasso applications lie somewhere between traditional and algorithmic methods in offering at least some hope of principled attribution.

To blame correlation between predictors for the lack of specificity seen in Table 3—where removing "top" predictors does not increase error rates—is only one possibility. Another is that a great many predictors may be weakly correlated with the response variable, and, because prediction is comparatively easy, these can be assembled into strong prediction rules in many different ways. A more attribution-friendly prediction algorithm would need, like the lasso, to be less promiscuous in selecting predictors. (I wondered if the early selections of the lasso-based skin cancer rule were strongly implicated?)

- (2) Both *War and Peace* and *The Big Bang* theory contain great writing, but they are great on different time scales. The actual big bang theory is a data-based estimate of the early universe. Physicists do not expect it to be the last word but it could predominate for decades or even centuries—not so, say, for the Netflix movie prediction algorithm. My point was not that prediction is inherently ephemeral; rather that current algorithmic predictions may tend by their construction toward a short shelf-life.
- (3) Sara Reardon's article in the February 2020 issue of *Scientific American* comments on a deep-learning pneumonia-predicting algorithm developed at Mount Sinai Hospital:

... It performed with greater than 90% accuracy on x-rays produced at Mount Sinai but was far less accurate with scans from other institutions. They eventually figured out that instead of just analyzing the images, the algorithm was also factoring in the odds of a positive finding based on how common pneumonia was at each institution—not something they expected or wanted the program to do.

The "clever Hans" effect might occur more subtly, and more widely than in the mammogram example.

- (4) There's no question that a preliminary use of say random forests is a good way to start a more traditional data analysis, as FHT recommend. (I have gotten fond of doing just that.) This does not mean that there are not differences worth understanding between traditional and algorithmic methods. The military and local police may work "hand in glove" in a disaster but they are still quite different organizations. The paper was intended to pinpoint the differences between the old and new data analysis methodologies, with the hope of encouraging improvements in both.

Section 8 of the paper repeatedly quotes Professor Cox's commentary on Leo Breimans' 2001 paper "Statistical Modeling: the Two Cultures," where he lucidly defends the traditional culture, the one Leo does not have much time for. His discussion here includes a tribute to the international nature of the statistics community—the happy fact that our field is not split into non-communicating polities. The possibility of such a split is a worrisome corollary of the two cultures point of view.

Parametric modeling, Criteria 4 in Table 5, is a hallmark of the best of traditional statistical theory. Fisher was the key figure in its centrality. At the same time, he argued against over-mathematization. Cox writes

An aspect of both Fisher and Bartlett's work, which I must admit I admire, is a total lack of concern with formal mathematical regularity conditions which, of course subject to due care, typically contributes little to understanding.

I could not agree more. Of course, Fisher and Bartlett combined their lack of concern with the ability to produce

profound results, a small club that includes Professor Cox as a member.

All of this is to say that in hoping for mathematical structure in support of the prediction algorithms, the wish is for more structure, not more math per se.

My thanks to the discussants for what seem to me to be an unusually interesting set of commentaries on a pressing topic. Under the best of circumstances, discussion papers put a strain on the Editor. These are not the best of circumstances (April 2020), but that has not stopped Regina Liu from putting all this together. Thanks Regina!